



**U.S. CENTERS FOR DISEASE
CONTROL AND PREVENTION**



Predicting Heart Disease Among U.S. Adults Using 2020 CDC Behavioral Risk
Factor Surveillance System (BRFSS) Dataset

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Abstract

According to the World Health Organization, heart disease is the leading cause of death globally and accounts for roughly a third of deaths worldwide. With rapid advancements in technology, there must be a way to harness the capabilities of modern computing to reduce the damage of heart disease around the world by attempting to find the leading causes of heart disease. In recent years, there have been attempts to do just that by using machine learning to predict the prevalence of heart disease in individuals based on specific medical data.

Our research involves using the 2020 CDC Behavioral Risk Factor Surveillance System (BRFSS). This data comes from the CDC's annual survey which involves telephone interviews in order to collect demographic, behavioral and health-related information about individuals across the United States. The BRFSS data set started with over 400 thousand observations and 30 factors. This was reduced to around 320 thousand observations and 17 health related predictor variables. Using this dataset, we wanted to find out if it is possible to predict the prevalence of heart disease.

Our first approach asked, "Which demographic, behavioral, and health-related factors are the strongest predictors of heart disease among U.S. adults?" We tested three different prediction models to see which would be the most accurate. The first model we used was logistic regression where we modeled the probability of heart disease as a function of the demographic, behavioral, and health-related factors. The second model used was the random forest model where multiple decision trees are constructed using random subsets of observations and predictor variables and then combined and analyzed to see if an individual would have heart disease or not. The third model used was a decision tree model where the data was partitioned into subsets based on

predictor variables and a structure of decision rules was made to classify individuals as having or not having heart disease. Based on the results from the tree, we can see which variables are found to be the most valuable for predicting heart disease.

We evaluated the predictive performance of the model with a stratified 10-fold cross validation. From these subsets, we learned the performance metrics such as AUC, accuracy and F-1 score. The results showed that across all models the strongest predictors were ‘Age Category’, ‘General Health’, ‘Diabetes’, and ‘Stroke’.

Our second approach asked; “What common characteristics define patient risk profiles and how do heart disease rates vary amongst these groupings?” We wanted to find out the attributes of individuals at high, medium, low, and no risk for heart disease. To do this, we decided to use k-means clustering to create these groups using the same demographic, behavioral, and health-related factors. We applied two methods: the elbow method and silhouette analysis. In the elbow method, we look at the graph of k-values to identify where there is no longer a significant difference and the graph starts to flatten.

To measure the performance of our second approach, we used chi-square testing and analysis of variance (ANOVA) to help us validate our k-means clustering analysis. Our analysis showed us two distinct groups of individuals: one group had older age, higher rate of heart disease, diabetes, stroke, and poor physical health and the other group had a younger age, lower rates of heart disease, diabetes, stroke, and better physical health. Both groups have some of the same attributes but the second group has a lower overall risk for heart disease.

This project is important because if our predictive modeling and clustering is accurate, it can be used in real life scenarios to predict prevalence of heart disease in individuals by identifying harmful traits and changing behaviors to minimize the risk of heart disease. It is not

guaranteed that our predictions will be correct every single time, but they are meant to act as a guideline.

Project Plan

Primary Company Details

Company Name: Centers for Disease Control and Prevention (CDC)

Address: 1600 Clifton Road NE, Atlanta, GA 30329

Company Communication:

Phone Number: 1-(800)-232-4636

Website: www.cdc.gov

[Email Contact Form](#)

Business Description:

The CDC is a federal agency and the leading public health institute in the United States. Its mission is to protect public health through disease prevention, promoting general health, and ensuring preparedness. The CDC aims to advance healthcare knowledge, science, and technology. They achieve results through data collection, identifying health trends, monitoring the spread of diseases, and influencing public health decision making.

Financials:

Latest Financial Data - September 30, 2025 (FY 2025)

Annual Budget -	\$9,683 (in Millions)
Assets -	\$1,154,693 (in Millions)
Liabilities -	\$1,154,693 (in Millions)
Net Cost of Operations -	\$1,887,206 (in Millions)
Net Position -	\$919,739 (in Millions)

Total Budgetary Resources - \$3,129,745 (in Millions)

No. of Employees - 11,814 (FY 2022)

Key Executives:

- Jay Bhattacharya, MD, PhD - Director of the Centers for Disease Control and Prevention
- Matthew Buzzelli - Chief of Staff for Centers for Disease Control and Prevention
- Joseph Uddo - United States Department of State, Chief Medical Officer

Major Competitors:

Because the CDC is a public government agency, it does not have competitors. Similar organizations include the National Institutes of Health and the World Health Organization.

Business/Analysis Opportunity:

By using statistical analysis and predictive modeling techniques on a dataset collected in 2020 in the United States with 319,795 observations and 17 indicator variables related to cardiovascular health, our team can determine how variables such as smoking, drinking, physical activity, and general health affect rates and severity of cardiovascular disease. The findings from such analyses can help with identifying high-risk individuals, improving prevention strategies, reducing healthcare costs, and improving overall public well-being.

Research Questions:

RQ1: Which demographic, behavioral, and health-related factors are the strongest predictors of heart disease among U.S. adults?

RQ2: What common characteristics define patient risk profiles and how do heart disease rates vary amongst these groupings?

Hypothesis

H1: Older age, poor general health, diabetes, stroke history, alcohol and tobacco usage, difficulty walking and lack of physical activity will be the strongest predictors of heart disease amongst U.S. adults.

Cardiovascular risk factors established in public health and medical research overlap with the variables presented in the hypothesis. According to the CDC, various health conditions, lifestyle habits and age contribute to increasing the risk of developing heart disease. As age increases, the cardiovascular system deteriorates over time thus putting more stress on the heart. Poor general health can lead to other conditions such as diabetes and increased stroke risk. Excessive alcohol and tobacco consumption can increase blood pressure levels and damage the heart and/or blood vessels. Difficulty walking and lack of physical activity can lead to weight gain and other medical conditions which all compound into higher heart disease risk. Demographic, behavioral and health-related factors are expected to be the strongest predictors of heart disease among U.S. adults utilizing predictive analysis and logistic regression.

H2: Unique patient risk profiles characterized by demographic, behavioral, and health-related factors will result from cluster analysis and the prevalence of heart disease will vary significantly amongst each profile.

With the removal of heart disease as a data point, we can utilize cluster analysis to group similar demographic, behavioral and health-related variables into distinct patient risk profiles/clusters. Factors such as age, BMI, diabetes status, stroke status, general health status, physical activity levels, alcohol and tobacco consumption, and other chronic health conditions are expected to contribute to developing these risk profiles. Clusters characterized with older age, more chronic conditions and unhealthier lifestyles are expected to show higher rates of heart disease than one with younger age, less chronic conditions and healthier lifestyles. Determining these unique patient risk profiles through cluster analysis will provide insight as to which combination of factors will influence heart disease in U.S. adults the most.

Data

The Indicators of Heart Disease dataset from Kaggle is derived from the 2020 CDC Behavioral Risk Factor Surveillance System (BRFSS). The BRFSS dataset was conducted by the CDC with annual telephone surveys and consists of over 400,000 interviews with U.S. adults and is one of the largest health survey systems in the United States. This dataset contains demographic, behavioral and health-related variables associated with heart disease risk, such as BMI, smoking status, alcohol status, age, sex, physical activity status and other chronic health conditions. The original 2020 BRFSS dataset contained over 300 variables and was reduced to approximately 40 variables in this Kaggle version. This condensed dataset has the most relevant factors for heart disease prediction and makes it more suitable for data analysis and predictive modeling of heart disease occurrence among U.S. adults.

Measurements

Heart disease serves as the dependent variable and is categorized as binary with a simple yes or no within the dataset. Independent variables can be categorized into demographic variables (sex, age, race), behavioral variables (smoking, alcohol consumption status, physical activity status, amount of sleep) and health-related variables (BMI, stroke status, bad physical health status in the past 30 days, bad mental health status in the past 30 days, difficulty in walking, diabetes status, general health status, asthma status, kidney disease status, skin cancer status).

Methodology

For research question #1, we are trying to determine which of the various demographic, behavioral, and health-related factors from the dataset are the strongest predictors of heart disease among U.S. adults. Heart disease is set as a binary value in the dataset as a Yes/No, which will be later converted into 1/0 as we clean the data. Multiple logistic regression will be utilized to determine each predictive factor's significance and relative importance and this will allow us to determine which specific factors contribute to the likelihood of heart disease.

For research question #2, we are trying to determine various patient risk profiles and how heart disease risk differs amongst those profiles. We will utilize cluster analysis after we remove the dependent variable, heart disease, on demographic, behavioral, and health-related characteristics. With the clusters established we can then compare and contrast the prevalence of heart disease across these groups to determine whether specific patient risk profiles are connected with higher disease risk.

Computational Methods and Outputs

For research question #1, we believe that utilizing a correlation heatmap will be the first step of our analysis to help visualize the relationship between heart disease prevalence and the predictor variables. Next, a logistic regression coefficient table will help determine the statistical significance of each predictor and their estimated effects. The confusion matrix will give us the number of true positives, true negatives, false positives and false negatives which helps us determine the model's accuracy and predictive capability. An odds ratio plot will then be utilized to help interpret the results from the coefficient table in a cleaner fashion. In addition, an odds ratio plot will complement the heatmap. Next, we will also create a ROC/AUC curve after our exploratory analysis and logistical regression to help visualize and determine the performance of the model and its predictive accuracy.

For research question #2, to determine the value of K, or the number of clusters, we will utilize both the elbow method and the silhouette score to help determine the most meaningful patient risk factor groups. Next, we will utilize K-Means to perform cluster analysis by training the model with a ratio of 8:2 of training:validation which will produce cluster assignments and centroids. We can then visualize the heart disease risk per cluster by a bar chart. A cluster heatmap will also be created to help identify any trends or patterns among the clusters. Lastly, we can utilize a chi-square test of independence to determine whether the heart disease risk across similar clusters are statistically significant or not.

Analysis Implementation

According to the CDC, heart disease “is the number one killer of both men and women” in the United States and is associated with various demographic, behavioral, and health-related risk factors. By identifying the strongest predictors of heart disease, this data could be utilized in order to help create awareness and to target prevention movements for individuals at risk of developing heart disease. While we may not have a cure-all medication to eradicate heart disease, the best medicine is simply prevention. The results of our first research question can help identify these risk factors. Our second research question focuses on identifying the patient risk profile rather than individual risk factors as heart disease can be a culmination of various conditions and lifestyle choices which all could compound together into other chronic health conditions. Understanding the patient risk profile can help organizations and communities better identify which combination of these risk factors contribute the most to cardiovascular issues.

Literature Review

Cardiovascular disease, more commonly known as heart disease, is the leading cause of death worldwide according to the World Health Organization, making it a vital area of research for medical professionals. In the United States alone, one person dies of heart disease every 34 seconds. There are many types of cardiovascular disease, but the main types consist of coronary artery disease, arrhythmias, vascular disease, and congenital disease.

Researching the factors associated with cardiovascular disease aid in prevention, diagnosis, and treatment. By analyzing behavioral and medical risk factors such as consumption of drugs and alcohol, physical activity, and obesity, researchers and medical professionals are better able to understand how and why cardiovascular disease develops. In addition, studying large datasets allows researchers to identify underlying trends and relationships that may be missed in clinical observations.

Data-driven research has an increasingly important role in protecting global health by improving treatment plans, prevention methods, and public health recommendations. Identifying individuals at a high-risk for developing severe cardiovascular complications can aid in reducing hospitalization rates and improving quality of life. By using statistical analysis and predictive modeling, researchers can identify which factors have the biggest impact on cardiovascular disease developing or worsening. Cardiovascular disease research helps in advancing predictive analytics and early intervention strategies, as well.

The information reviewed on the CDC website ranks Ischaemic heart disease as the number 1 cause of death in the United States. It also identifies several major risk factors associated with cardiovascular conditions that are factors in the dataset we are using. Building on these findings, this study will analyze the heart disease dataset to determine how demographic

and overall health related variables are associated with the presence of heart disease. We will also use a cluster analysis to focus on identifying patterns, relationships, and potential predictors within the dataset to better understand factors that may contribute to cardiovascular risk.

While the WHO and CDC focus on population health trends, our predictive model will mainly be focusing on predicting heart disease risk for individual patients. Additionally, in the Leading causes of death website, when looking at the United States, there are two different types of heart disease in the top 10 (WHO). At number 1 is Ischaemic heart disease, and at number 9 is Hypertensive heart disease. In our dataset, the heart disease indicator is binary, leading to the type of heart disease not being distinguishable. The UCI's Heart Disease dataset does not include any mental health factors in the dataset whereas the CDC Indicators of Heart Disease dataset includes both physical and mental health factors. In the CDC article *About heart disease and mental health*, it is discussed how many mental health factors such as anxiety, depression, and chronic stress lead to increased chances of heart failure and cardiovascular disease. This may lead to new insights that were not found in the UCI analysis.

A similar study was done (Babu, 2024) with clustering analysis using the UCI Heart Disease dataset. This study used the k-means clustering approach which initially places k centroids randomly in the feature space then each data point is put into a cluster based on the distance from one of the centroids. After the data points are assigned to clusters, the averages are calculated and the process repeats until there is no longer a significant change.

Similarly another study (Rimal et al., 2025) was done using four models (support vector machine, logistic regression, k nearest neighbors, and random forest) and the same UCI heart disease dataset to see which model would be the best at predicting the prevalence of heart disease

in a given patient. In this instance, it was found that the SVM model had performed the best out of the four.

Both studies show that machine learning techniques can be used to accurately predict the prevalence of heart disease. This is important because it can be directly applied to real life scenarios where patients can see what kind of risk they are at for heart disease. Application of this technology can help to save lives by addressing certain causes for heart disease.

Heart disease remains one of the most significant public health challenges worldwide as well as the leading cause of death in the United States (CDC). By analyzing heart disease datasets we can better understand the relationship between demographic and health-related variables associated with heart disease. Gaining a deeper understanding of the factors associated with heart disease may enhance predictive models, allowing healthcare professionals to identify high-risk individuals and implement preventive strategies to reduce the likelihood of complications from heart disease. The other studies reviewed demonstrate that machine learning approaches to predicting heart disease risk can effectively identify patterns in at-risk individuals. Overall, this study aims to contribute to a better understanding of the factors associated with heart disease and demonstrate how machine learning techniques can support early detection and prevention efforts.

Final Research Questions

RQ1 - Which demographic, behavioral, and health-related factors are the strongest predictors of heart disease among U.S. adults?

RQ2 - What common characteristics define patient risk profiles and how do heart disease rates vary amongst these groupings?

Exploratory Data Analysis

Understanding the risks associated with cardiovascular disease is essential for reducing healthcare costs, improving prevention strategies and public welfare, and identifying high-risk individuals. The purpose of this exploratory data analysis is to identify trends and patterns in the relationships between health factors such as smoking, drinking, age, and bmi with the presence of cardiovascular disease.

The data we are using comes from the 2020 Centers for Disease Control and Prevention Behavioral Risk Factor Surveillance System (BRFSS) Annual Survey which collects data on health conditions, health-risk factors, and preventative services. There are 319,795 observations in this dataset, with 17 health related predictor variables and a binary target variable:

HeartDisease: signifies if cardiovascular disease is present(1) or absent(0) in the subject

BMI: the subject's body-mass index

AlcoholDrinking: signifies if the subject consumes alcohol

Stroke: signifies if the subject has had a stroke

PhysicalHealth: measure of the subjects physical health from 0.0 - 30.0

MentalHealth: measure of the subjects mental health from 0.0 - 30.0

DiffWalking: signifies if the patient has difficulty walking

Sex: the subject's sex (male/female)

AgeCategory: the age range of the subject

Race: the race of the subject

Diabetic: signifies if the subject is diabetic

PhysicalActivity: signifies in the subjects is physically active

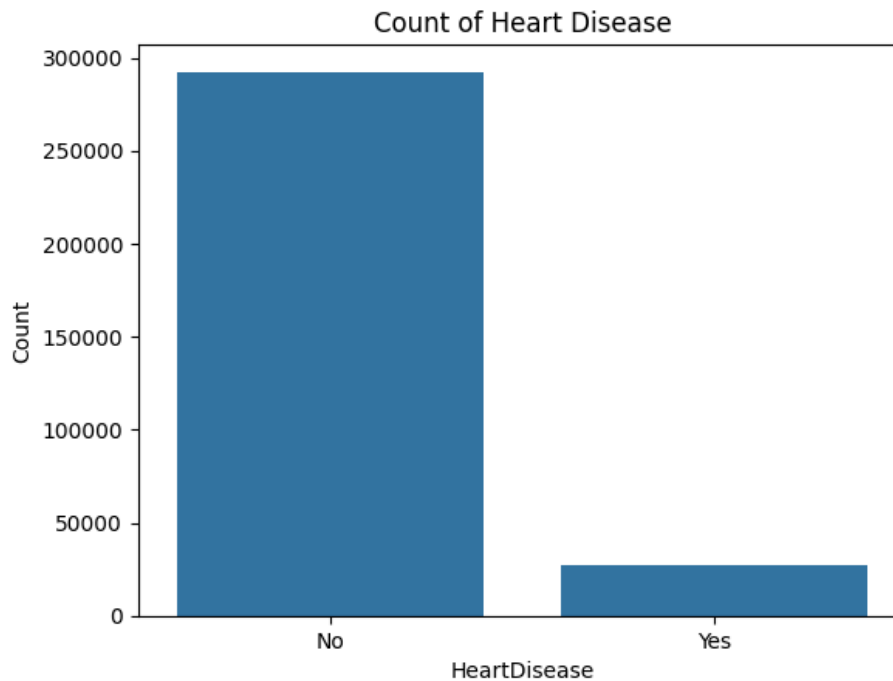
GenHealth: general health categories

SleepTime: average sleep time of the subject

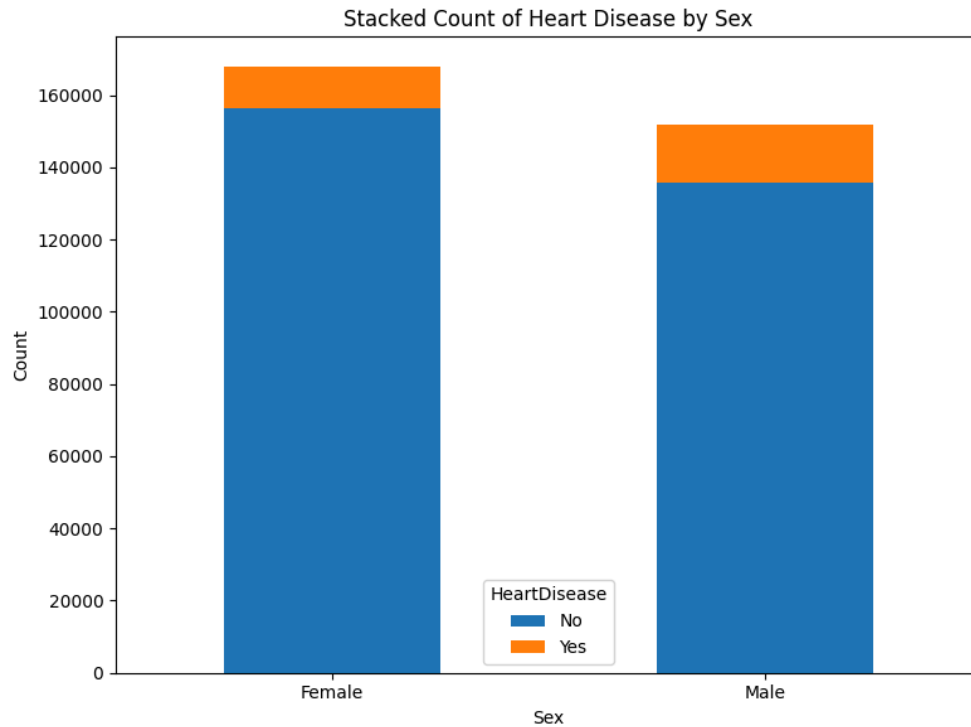
Asthma: signifies if the patient has asthma

KidneyDisease: signifies if the subject has kidney disease

SkinCancer: signifies if the subject has skin cancer

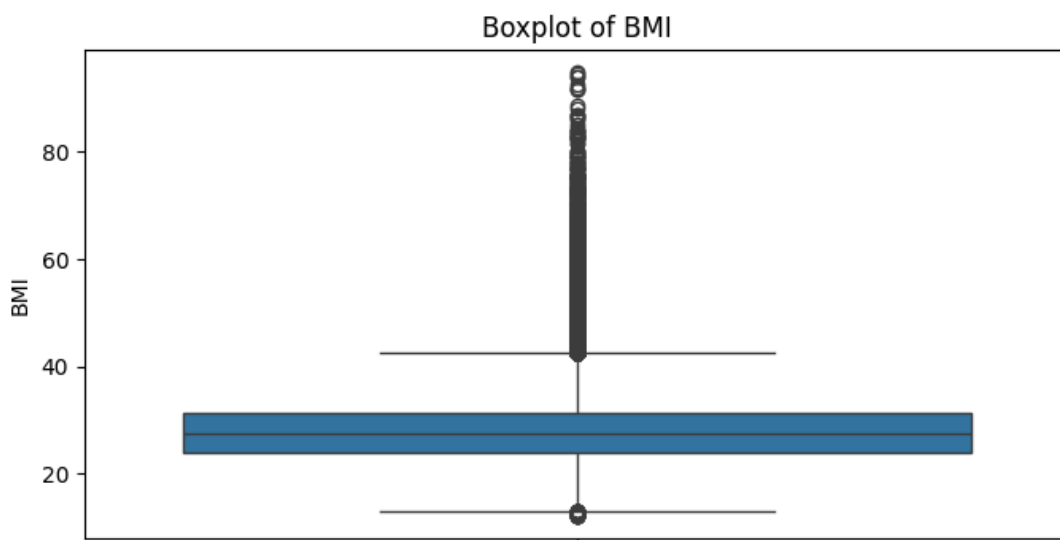
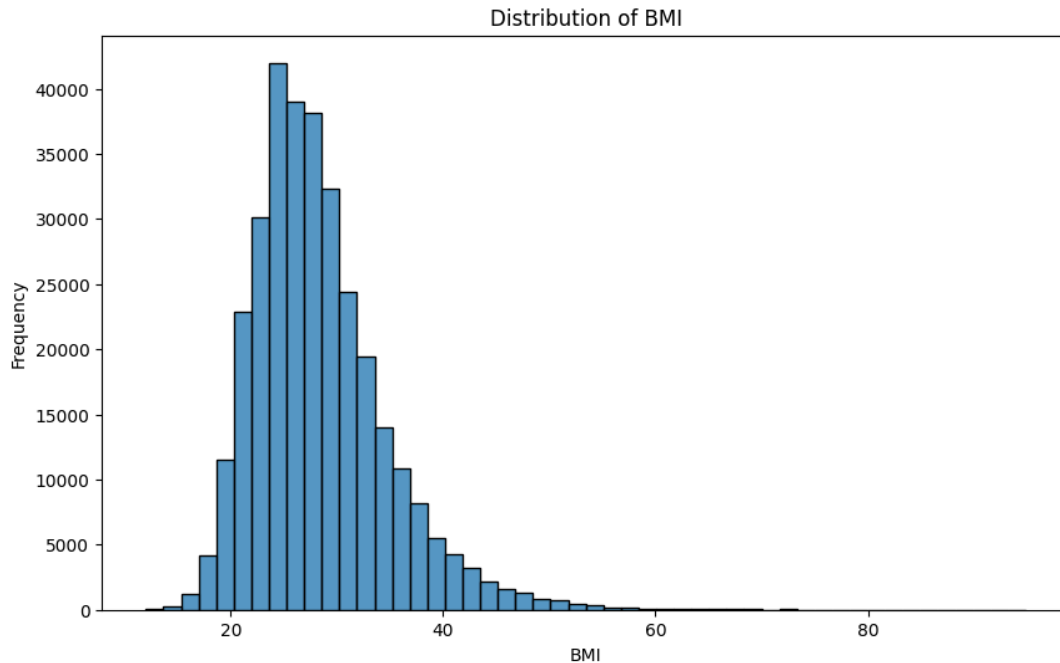


Because health issues may manifest differently for males and females, it is vital to understand the rates of cardiovascular disease in both sexes. Visually, it appears males have slightly higher rates of developing a cardiovascular disease.



Using a count plot, 292,422 of the individuals in the dataset do not have heart disease and 27,373 do. This means that out of the 319,795 individuals in the dataset 8.5% have heart disease. Of that population, 16139 are male and 11234 are female. Considering the size of the dataset it can be said that there is not a correlation between heart disease and sex as the numbers for both are close despite the males having slightly higher numbers of heart disease.

We then created a histogram and boxplot to visualize the distribution of body-mass index (BMI) among the dataset.

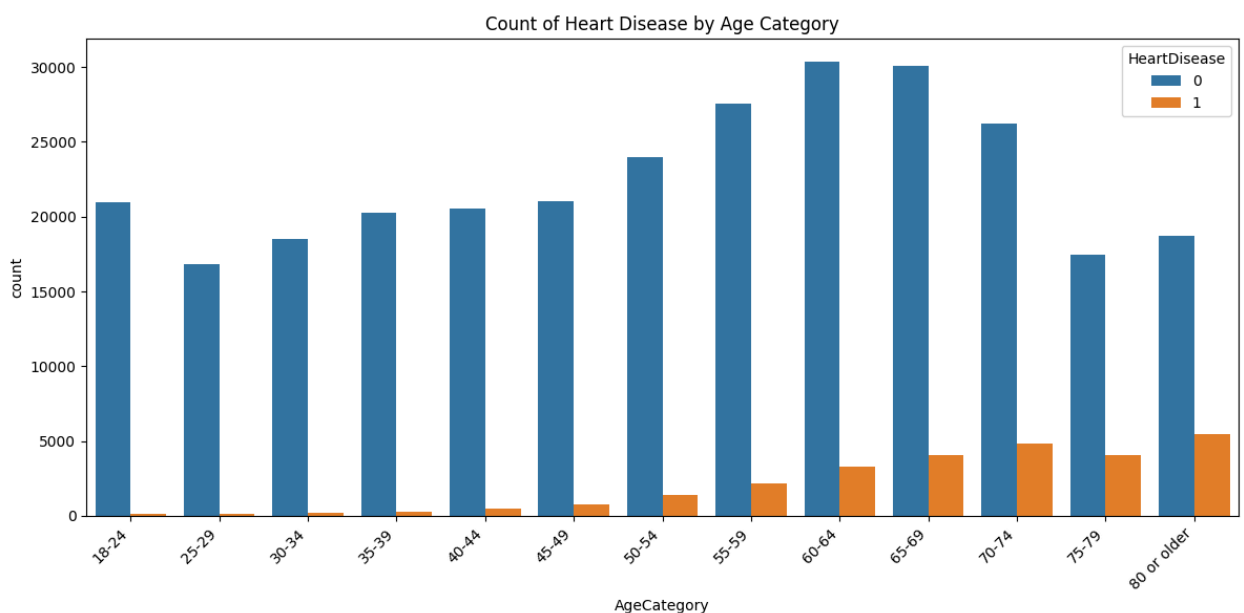


The median BMI in this dataset is approximately 26-29, indicating that the majority of the individuals in this dataset are considered medically overweight. The middle 50% of observations are in the 24-32 range. The distribution of BMI is positively skewed, with a noticeably high proportion of outliers having a higher BMI than the median. The minimum BMI observed in this dataset is 12.02, and the maximum is 94.85. The maximum BMI value is

virtually impossible for a living human to have, so there is some concern to be had with the integrity of this dataset. This is always a concern with self-reported data, so caution in analyzing these results is vital to the integrity of the findings made using this dataset.

Because obesity is considered a risk factor for cardiovascular disease, the findings from this analysis indicate the relationship between BMI and cardiovascular disease should be evaluated further. The visualizations demonstrate that the BMI of individuals in this dataset vary widely, which provides an opportunity to further analyze how varying BMIs affect the risk of developing cardiovascular disease.

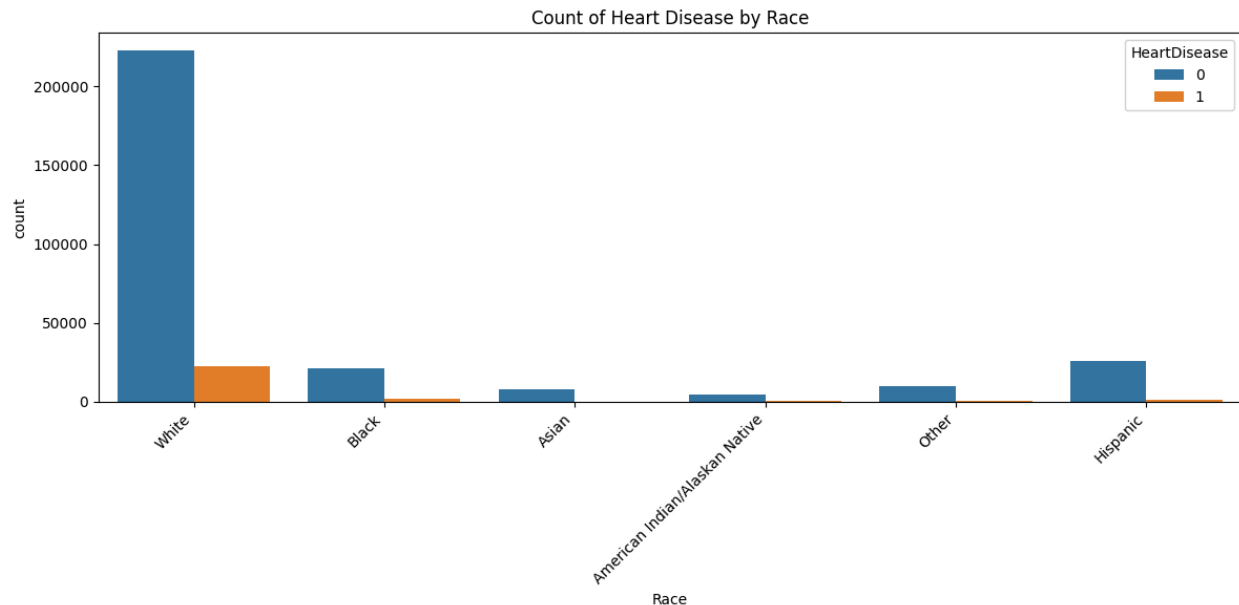
The distribution of cardiovascular disease by age was examined to determine whether age has a significant association with heart disease.



Individuals without cardiovascular disease make up the majority of this dataset. There is a clear trend in this dataset that suggests there is a positive correlation between age and developing cardiovascular disease. This supports current findings about cardiovascular disease.

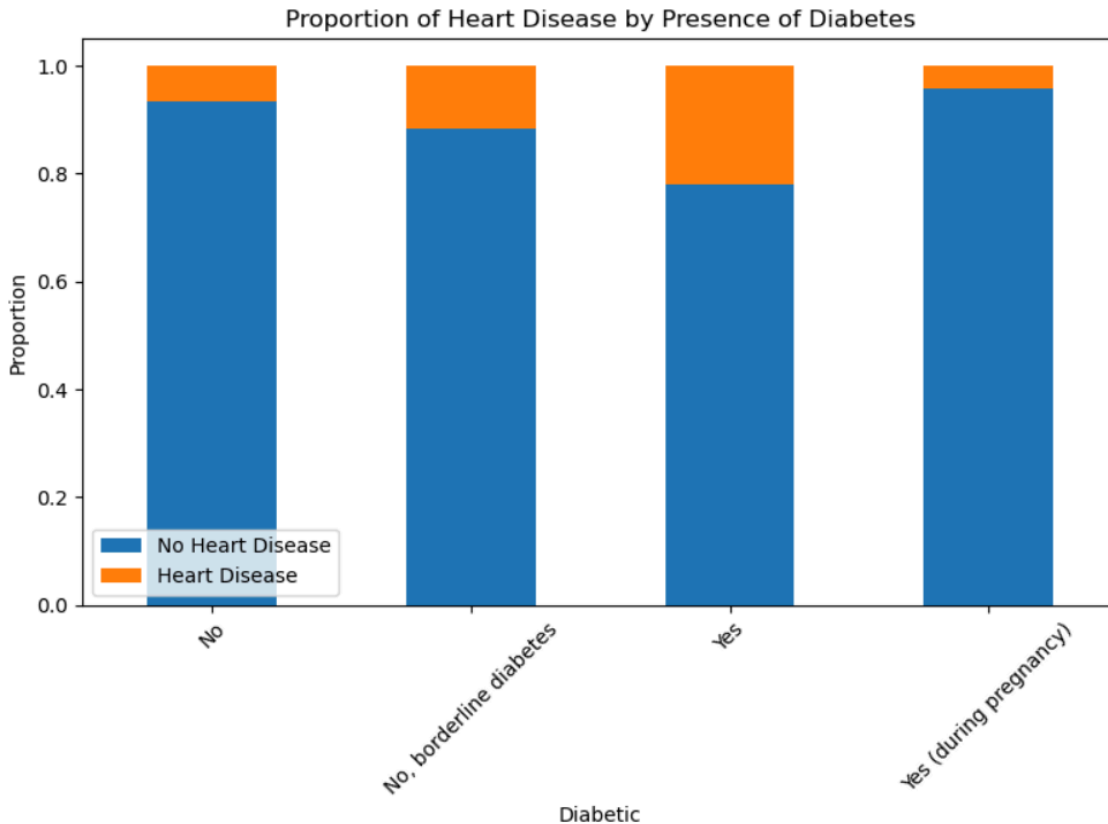
The noticeable increase in cardiovascular disease in older individuals indicates that age may be an important factor in predicting and assessing cardiovascular disease risk.

Next, we looked into how the presence of cardiovascular disease is distributed by race.



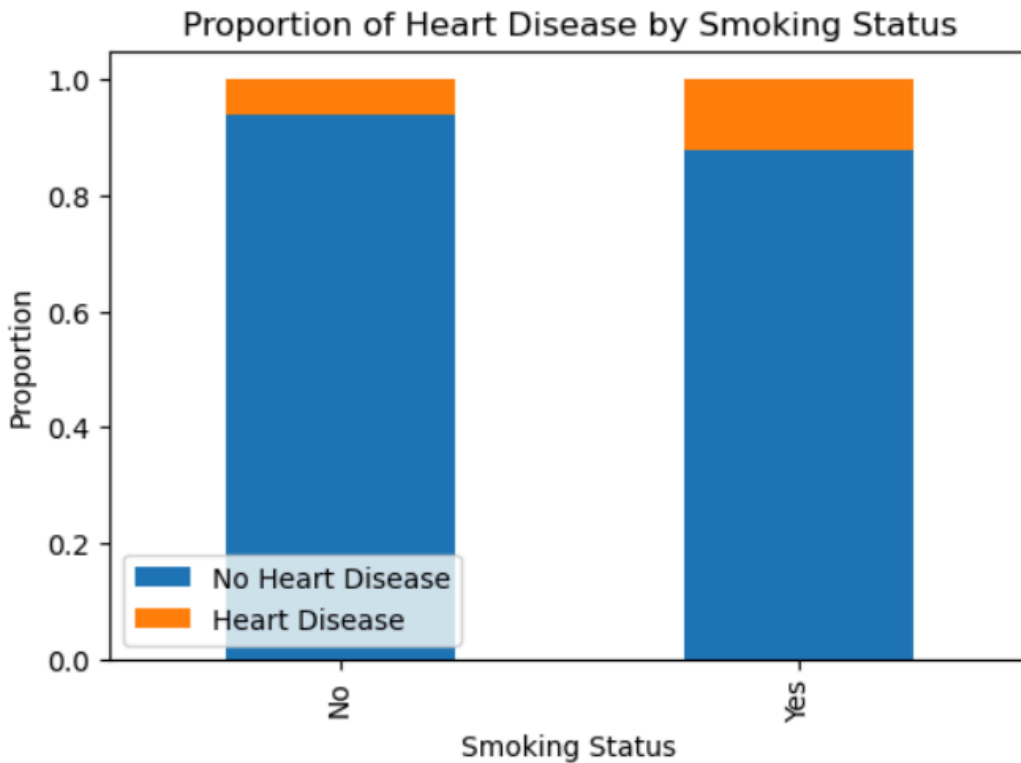
This visualization provides useful insight into the demographic distribution of this dataset. There is the presence of heart disease in every racial group, but further statistical analysis will be needed to determine if there is any meaningful correlation between race and prevalence of heart disease. The graph shows that White has the largest number of individuals with heart disease but White also accounts for roughly 76.7% of the dataset.

Analyzing preexisting health conditions can give insight into the development of cardiovascular disease. Diabetes is prevalent among Americans, so analyzing a possible link can be vital in cardiovascular disease prevention.



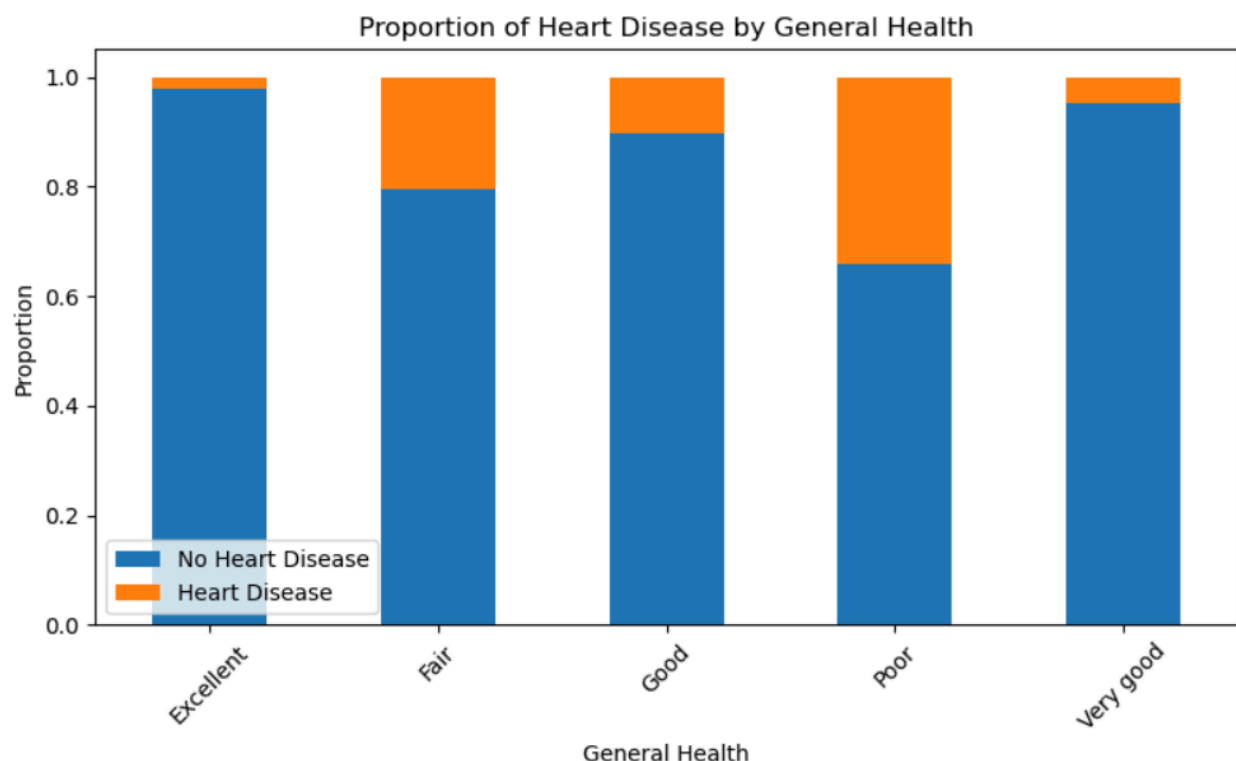
This visualization shows us the comparison of individuals who have heart disease and various categories of diabetes. In our dataset, 40,802 individuals have diabetes (12.8% of the total dataset) and 6781 individuals are at risk for diabetes (2.1% of the total dataset). Out of the individuals who have diabetes, 21.952% of those individuals have heart disease. This shows us that diabetes is not a strong predictor for heart disease but it does indicate a small factor.

This stacked bar chart demonstrates the proportion of smokers and non-smokers and the prevalence of cardiovascular disease. The smokers category indicates a clearly larger proportion of individuals with cardiovascular disease.



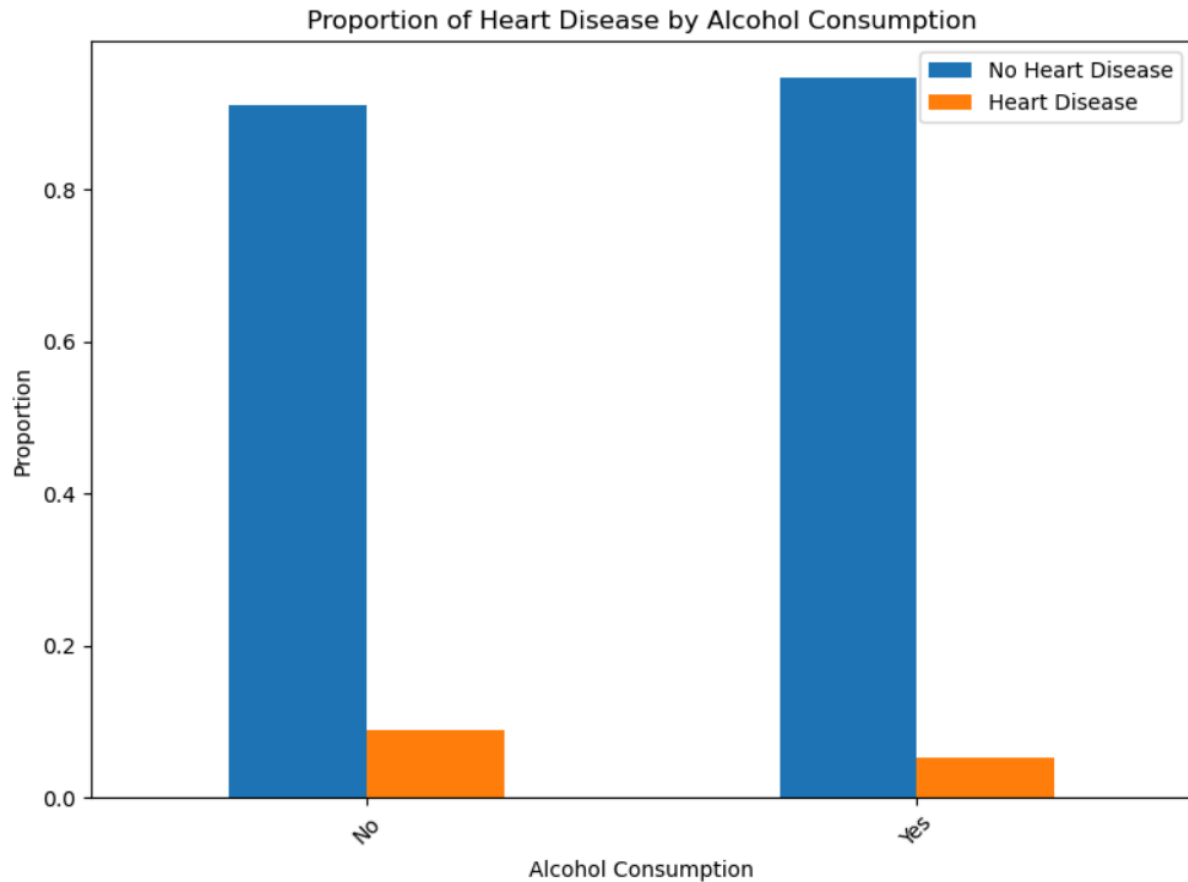
There are 11,336 non-smokers (6.0% of non-smokers) and 16,037 smokers (12.2% of smokers) with a positive diagnosis for cardiovascular disease. These findings suggest a positive correlation between smoking and developing heart disease. The proportion of heart disease cases for smokers was double that of non-smokers, which indicates that individuals who smoke are at a higher risk for developing cardiovascular disease. This notion is consistent with current medical findings on the link between smoking and developing cardiovascular issues. This chart suggests the link between smoking and cardiovascular disease risk should be further evaluated because smoking status may be a significant factor in predicting cardiovascular disease risk.

A visualization showing the rates of cardiovascular disease for individuals with differing levels of overall general health is important to understanding the full picture of cardiovascular disease risk.



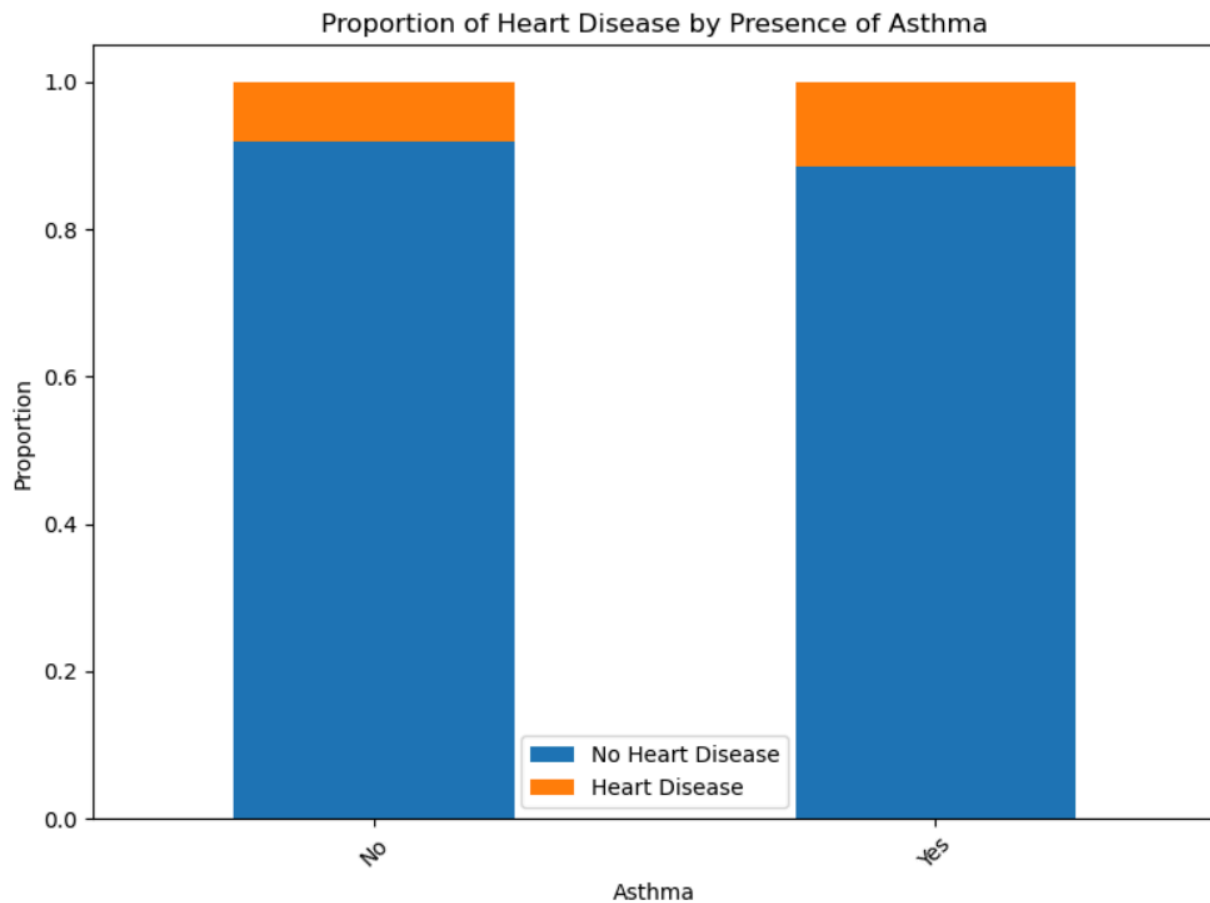
There is a distinct increase in the prevalence of cardiovascular disease with individuals who have fair or poor health. Approximately 34.1% of individuals with poor health and 20.4% with fair health report having cardiovascular disease. This is a stark difference compared to the 2.2% of individuals with excellent health and 4.7% with very good health that report having cardiovascular disease. These results show a noticeable pattern in the data: as overall health declines, the prevalence of heart disease increases. This suggests there is a strong correlation between overall health and the presence of heart disease.

Next, we took a look at how the rates of cardiovascular disease vary based on their status of alcohol consumption.



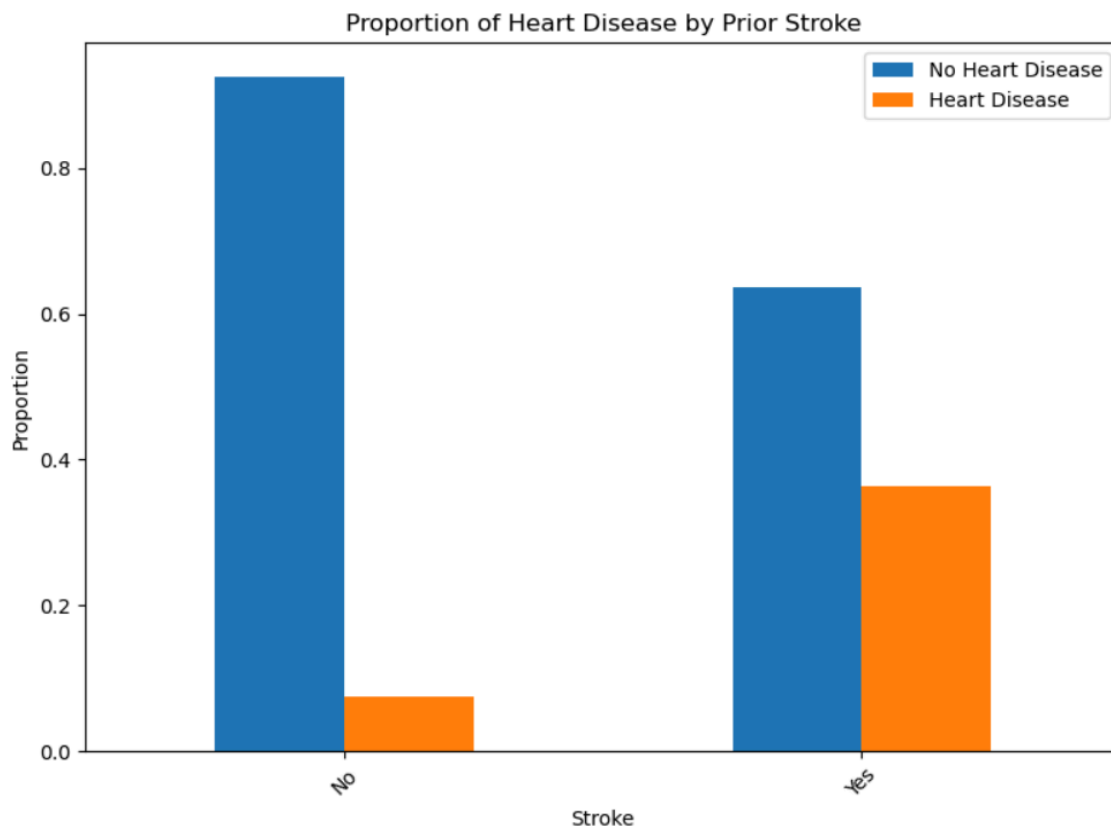
Approximately 8.8% of respondents who do not drink alcohol reported a positive cardiovascular disease diagnosis. Respondents who drink alcohol reported having a lower rate of cardiovascular disease with 5.2%. The lower rate in cardiovascular disease suggests being looked into. There could be a variety of reasons for this, such as there not being a distinction between moderate and heavy consumption or those already diagnosed with a cardiovascular disease may avoid alcohol. While drinking alcohol is associated with the lower rate of cardiovascular disease, this does not suggest a positive correlation between alcohol consumption and preventing heart disease. The results associated with the links between alcohol consumption and developing heart disease should be examined carefully so as to not overlook any other factors that may be contributing to this.

Looking at the barchart displaying the prevalence of heart disease in individuals with asthma, there is a noticeably larger proportion of individuals with heart disease that have asthma compared to those who do not.



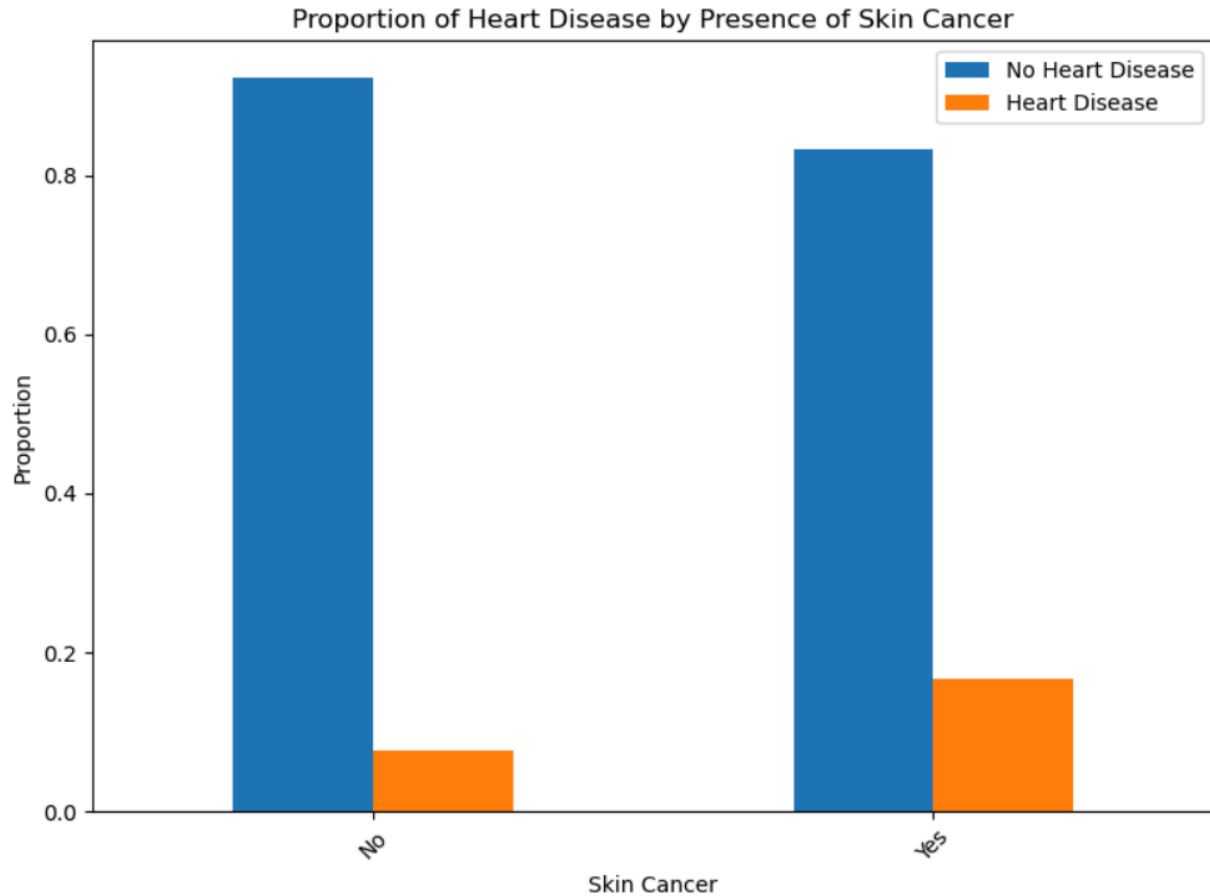
Having preexisting health conditions increases your risk of developing cardiovascular disease, so it is important to analyze the relationship between asthma and heart disease. Approximately 11.5% of individuals with asthma have a cardiovascular disease compared to the 8.1% of individuals without asthma. There seems to be a slight, but not significant, correlation between the two diseases, and the correlation may be explained by a combination of other factors.

The barchart displaying the presence of a prior stroke visually indicates a strong relationship between the presence of a cardiovascular disease and having had a stroke.



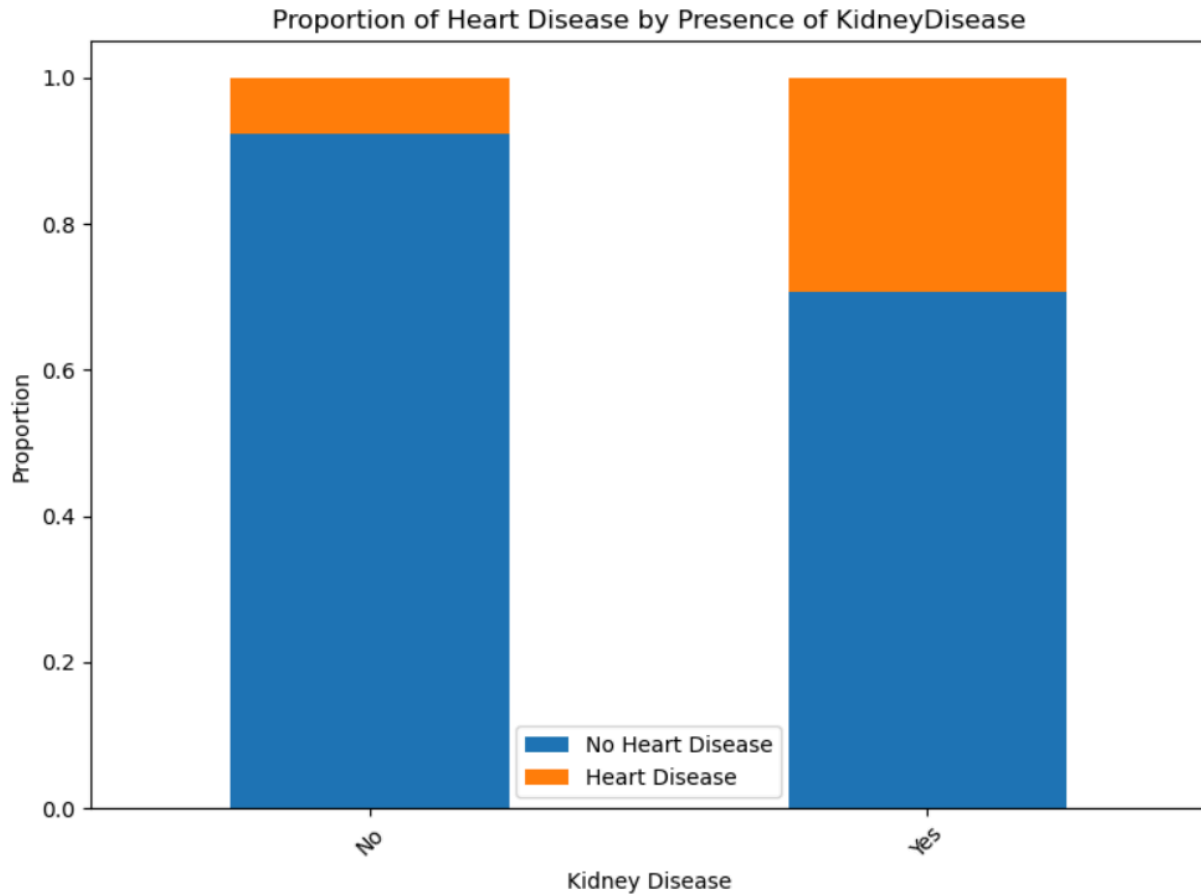
One of the biggest factors for looking out for heart disease is whether or not an individual has had a prior stroke. Approximately 3.8% of individuals in our total dataset have experienced a stroke and 36.4% of those individuals have heart disease. Although there is definitely some strong correlation between having a stroke and heart disease, 7.5% of individuals who have not had a stroke still have heart disease.

Individuals who experience skin cancer have a much higher rate of cardiovascular disease (16.7%) compared to those who do not have skin cancer (7.7%). This is clearly demonstrated in the graph below.



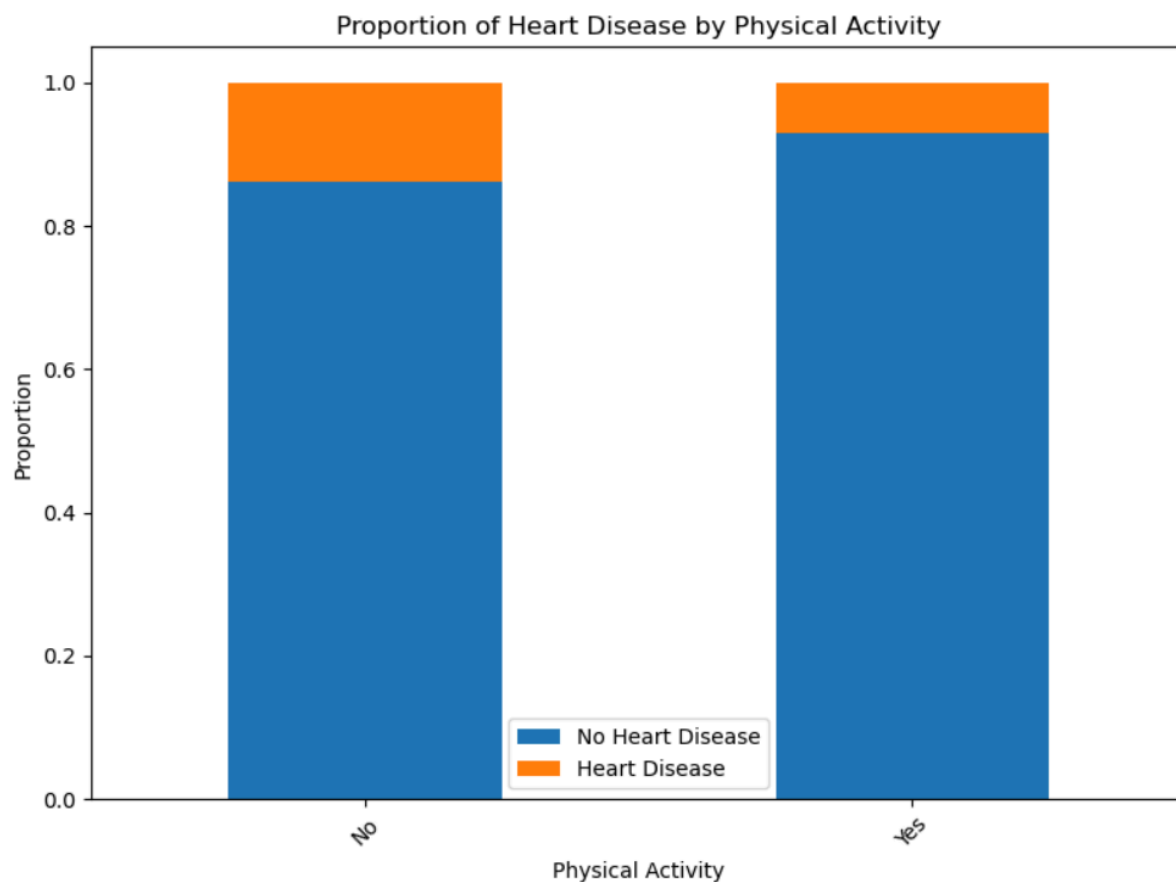
The results of this barchart indicates that there may be a relationship between the presence of skin cancer and the risk of developing a cardiovascular disease, as the rate of cardiovascular diseases among individuals with skin cancer is over twice that of those without. There may be underlying factors associated with this, such as the fact that older individuals have higher rates of cardiovascular disease and likely also have higher rates of skin cancer.

The presence of kidney disease also seems to have a strong relationship with the presence of cardiovascular disease.



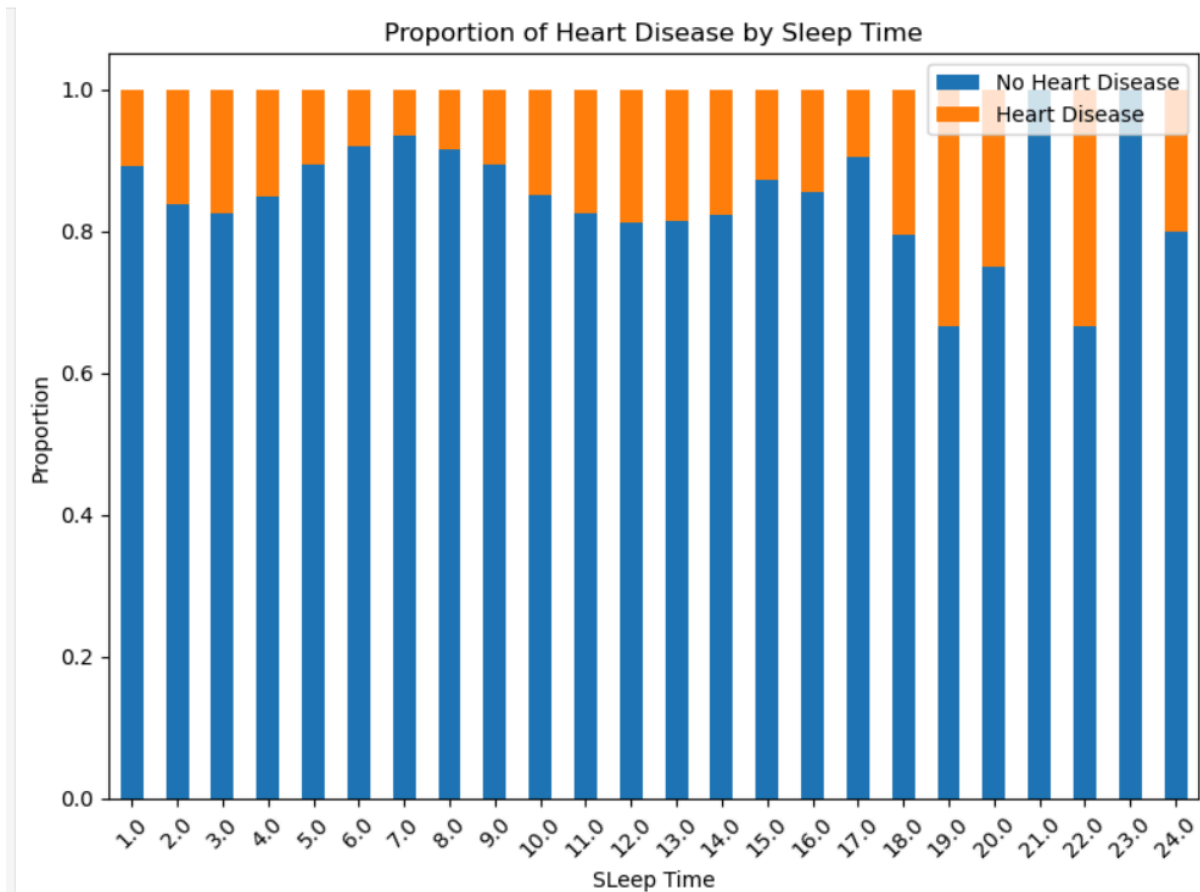
This stacked bar chart compares the proportion of heart disease among individuals who do and do not have kidney disease. Individuals without kidney disease have a relatively low presence of heart disease whereas individuals with kidney disease show a substantially higher prevalence of heart disease. This suggests a strong positive association between kidney disease and heart disease.

A higher amount of physical therapy is associated with better overall general health, so understanding its link to cardiovascular disease is vital. Based on the initial look, it is clear that there is a higher rate of cardiovascular disease in individuals who are not physically active compared to those that are.

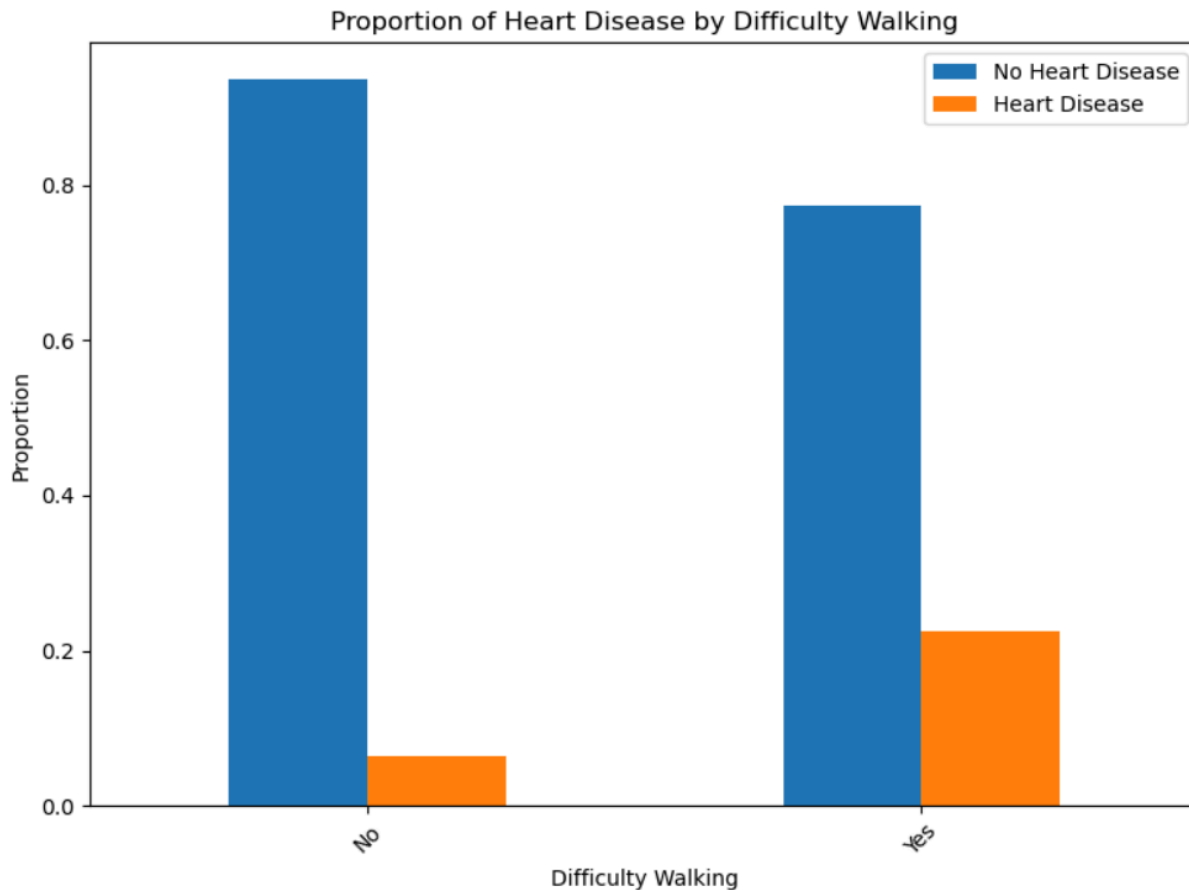


With approximately 13.8% of individuals who do not exercise regularly having cardiovascular disease, the rate is almost double that of those who do exercise regularly, 7.0%. These results suggest that those who are more physically active may be less likely to develop a cardiovascular disease. This does not automatically indicate a direct correlation because there may be other factors influencing the prevention of heart disease, such as physical activity being linked to better overall health.

Upon initial investigation, there is a slight pattern in the correlation between the amount of time sleeping and having a cardiovascular disease. The rates of heart disease vary widely between the range of hours slept. There is an up and down pattern throughout the range of sleep times, with the highest rates being in those who sleep for longer hours at a time.



The lowest rate of cardiovascular disease was at a sleep time of around 6-8 hours at approximately 6.5%. Individuals who sleep less than 4 hours or more than 10 hours seem to have higher rates of heart disease with rates of approximately 15-20%.



Difficulty walking is closely associated with physical health and this visualization shows that those who do have difficulty walking are more likely for heart disease. This factor is somewhat broad and can be very different for many cases such as a lost limb, old age, or an obesity problem. Out of the total dataset, 44,410 identified as having difficulty walking (13.9%). Of those individuals 22.58% have heart disease compared to 17,345 individuals who do not have difficulty walking but do have heart disease (5.424% of the total dataset). This shows us that there is somewhat of a correlation between difficulty walking and heart disease but it is not a strong factor.

Overall, this exploratory analysis revealed significant insight into the dataset. Several relationships were revealed between different health factors and the presence of a cardiovascular disease. Individuals who are smokers, have preexisting conditions, are older in age have higher rates of heart disease. Similarly, there are lower rates of heart disease linked with higher physical activity and better overall health. While these findings are purely descriptive, these factors are worth looking into as studying the relationship between the variables can be useful in cardiovascular prevention and treatment.

The observed relationships may have compound effects such as age, overall health, and lifestyle factors. The data will be useful in learning more about the links between health factors and cardiovascular disease through statistical and predictive modeling.

Methodology

RQ1 - Which demographic, behavioral, and health-related factors are the strongest predictors of heart disease among U.S. adults?

Heart disease, a binary variable in our dataset, will serve as the target variable for our analysis. Predictor variables will come from three categories such as demographic, behavioral, and health-related factors. Demographic variables include AgeCategory, Race, and Sex. Behavioral factors include Smoking, AlcoholDrinking, PhysicalActivity, SleepTime, PhysicalHealth and MentalHealth. Lastly, health-related predictors include BMI, Stroke, DiffWalking, Diabetic, Asthma, KidneyDisease, and SkinCancer. The main objective of our analysis is to determine which variables will contribute the most towards the likelihood of heart disease among U.S. adults.

In order to prepare our data, we need to sort our dataset into binary, ordinal and nominal categories. For binary variables, we need to convert the “Yes” and “No”, into 1’s and 0’s, respectively. Age is categorized in brackets, such as 18-24, 25-29, 30-34, etc. up until 75-79 and lastly, 80+. General health is labeled similarly within the dataset such as excellent, very good, good, fair and poor. Both of these ordinal variables can be assigned a number from 1-13 age groups and 1-5 (from poor to excellent). One-hot encoding will be utilized to transform nominal variables such as race into multiple separate binary columns for each specific race. This will allow us to utilize each racial category as a reference group for regression analysis without the influence of artificial rankings. The continuous variables, such as BMI, SleepTime, PhysicalHealth and MentalHealth are already numeric and do not need further manipulation. For outliers, we plan to examine the continuous variables of the dataset through boxplots and Interquartile Range (IQR). These two methods will help identify any outliers in the dataset to

determine whether or not they are valid data entries or errors or medically plausible observations. It is important to distinguish the difference between them because outliers that are medically plausible should be retained in order to provide accurate and meaningful heart disease predictions. For example, if one person's height is 5'7" with a BMI of 100, they would weigh approximately 639 pounds. This would be an extreme outlier if the upper range of the IQR of BMI is 50 but completely plausible as the world record of the highest BMI recorded in history is 204.

We plan to validate our three model's accuracy and effectiveness through stratified 10-fold cross-validation. Stratified 10-fold cross-validation is a method where the dataset is divided into 10 equal parts, or "folds". The model is then run ten times with stratification which preserves the same distribution, for example, each fold will contain 85% no and 15% yes for heart disease. The purpose of stratification is to preserve proportions of every subdivided dataset and this is very important because in healthcare data, the target variable of heart disease is typically a small portion of the entire dataset. In each of these ten runs, nine folds are utilized to help train the model and the last fold is used to test the model's performance. The process being repeated ten times allows for each fold to serve as the test set exactly once. With each of our three models having ten folds, we can then compare the area under the ROC curve value (AUC) to determine which models will have the strongest predictive performance.

Modeling techniques:

- 1) Logistic Regression

- Model the probability of heart disease as a function of the demographic, behavioral, and health-related predictors that will estimate the coefficients for

each predictor variable to demonstrate how the log-odds of having heart disease are determined by changes in the independent variables.

2) Random Forest

- Construct multiple decision trees using random subsets of observations and predictor variables, then aggregate their predictions to classify individuals as having or not having heart disease. Importance measures will assess the contribution of each predictor, determining the factors most influential in predicting heart disease.

3) Decision Tree

- Partition the data into subsets based on predictor variables, creating a structure of decision rules that classify individuals as having or not having heart disease. The resulting tree will be analyzed to identify the variables and decision paths that are most important for predicting heart disease.

RQ2 - What common characteristics define patient risk profiles and how do heart disease rates vary amongst these groupings?

K-Means clustering will be utilized to identify the patient risk profiles based upon demographic, behavioral, and health-related factors. This analysis will aim to determine whether heart disease prevalence differs amongst these profiles based on shared similar characteristics. Heart disease as a variable will be removed prior to clustering to help ensure that clusters are formed only from patient attributes instead of disease outcome.

As mentioned above for RQ1, the data preparation consists of converting binary variables from “Yes” and “No” to “1” and “0”, respectively. Ordinal variables will be converted into an

ordered numerical scale and nominal variables will be transformed with one-hot encoding. The continuous variables such as BMI, hours of sleep, days of bad physical and mental health will need to be standardized for cluster analysis as the ranges of these variables differ. Sleep hours range from 0 to 24, physical and mental health range from 0 to 30 and BMI ranges from 12.02 to 94.85 according to our EDA. We will utilize z-score normalization in this scenario for the continuous variables to prevent BMI (the variable with the highest range) from disproportionately affecting the formation of these clusters. Z-score standardization will help rescale variables relative to their respective means and standard deviations.

In order to determine the optimal K for clustering, or the amount of optimal clusters, we will utilize both the Elbow method and Silhouette Analysis. The Elbow method consists of first determining a range of K values, such as 1 through 10, to be run through K-Means in order to calculate the Within-Cluster Sum of Squares (WCSS). The WCSS shows the distance between the cluster centroids and the data point. A plot of K on the X-axis vs. WCSS on the Y-axis will then be graphed and the point where the curve starts to flatten is the “elbow” or the optimal K. The Silhouette method is a process utilized to determine the quality of the clustering solution by measuring the distances of the data points between all the other points within the same cluster relative to neighboring clusters. The output of this process results in a silhouette score (ranging from -1 to +1) assigned to each K value to determine the optimal K. This will help validate the result from the Elbow method to ensure the correct number of patient risk profiles are meaningful groupings within the population.

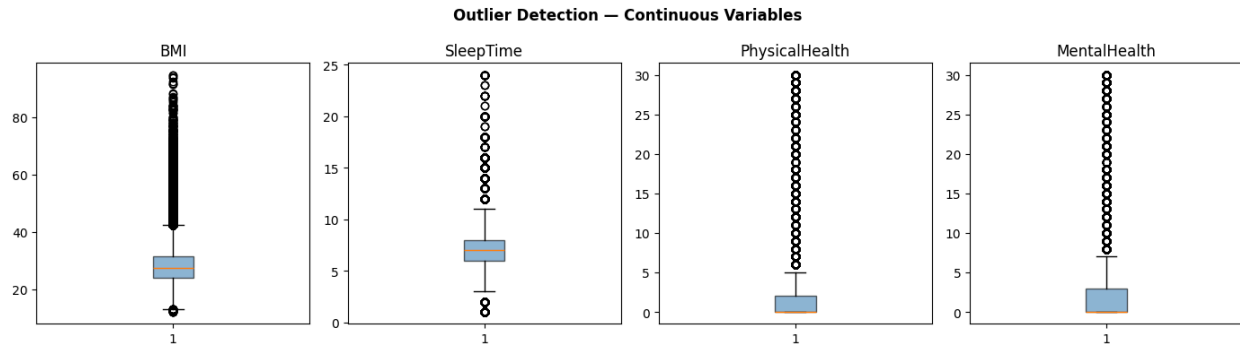
Modeling techniques:

- 1) K-Means Clustering

- Assign individuals to k distinct clusters based upon similarities across selected variables. Cluster centroids and summary statistics will be used to characterize patient profiles, and heart disease rates will be compared across clusters to assess disease prevalence among the differently identified risk groups. The algorithm will repeat until data points no longer switch clusters, or until a maximum number of clusters is reached.

Data Visualization

Outlier Analysis / IQR

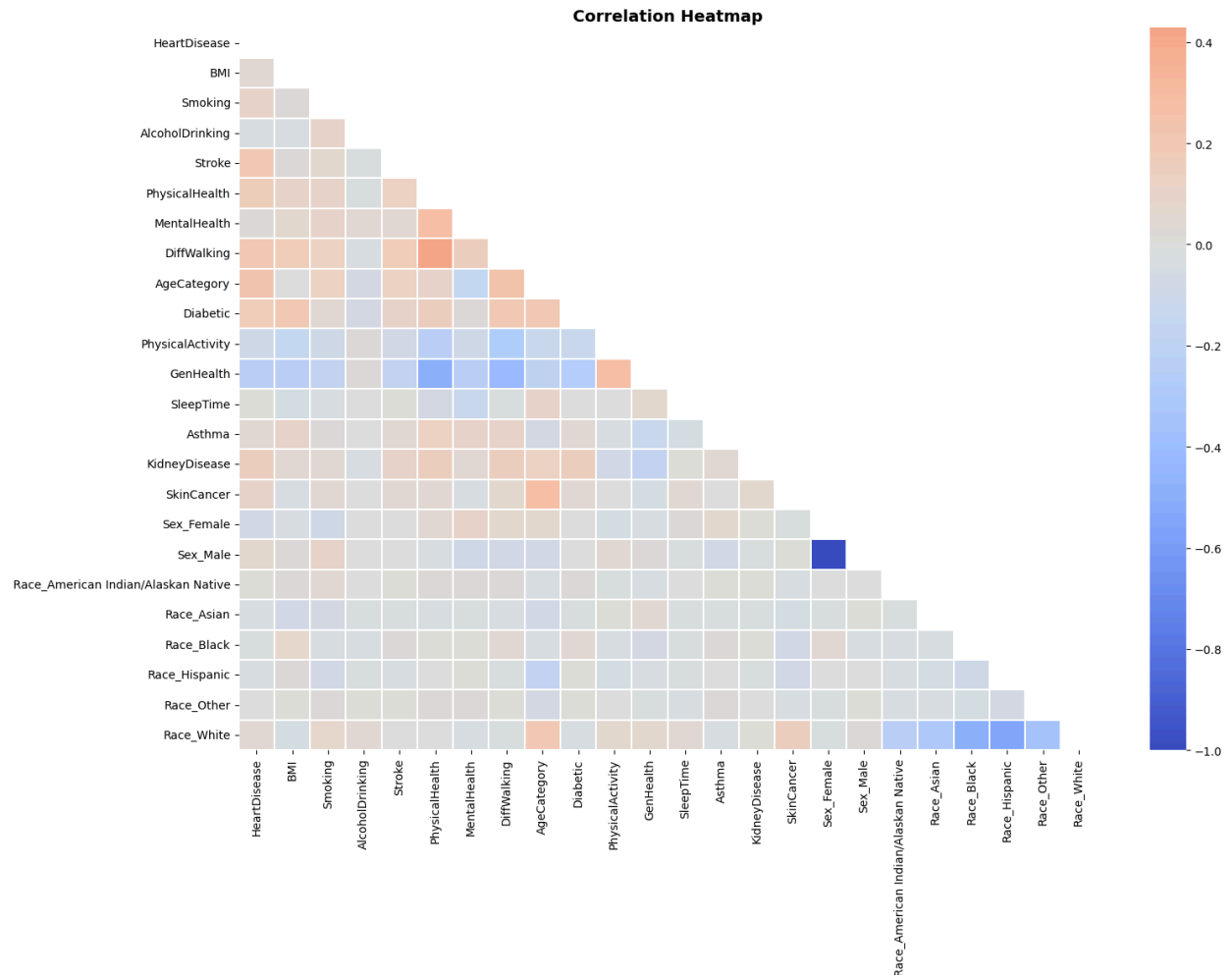


	IQR - Lower Bound	IQR - Upper Bound	Outliers Flagged
BMI	12.9	42.5	10,396
SleepTime	3.0	11.0	4,543
PhysicalHealth	-3.0	5.0	47,146
MentalHealth	-4.5	7.5	51,576

The boxplot visualization and Interquartile Range (IQR) method were both utilized in tandem to identify any potential outliers amongst the continuous variables: BMI, SleepTime, PhysicalHealth and MentalHealth. The importance of recognizing outliers in regards to medical-based data is determining invalid data versus extreme but legitimate data. While the IQR method depicted a large number of observations for outliers in both PhysicalHealth and MentalHealth, we can evaluate that the respondents either reported very few days (such as zero) or a large number of unhealthy days (such as 30 or everyday of the recorded month). Due to the

consideration of impactful medical data, the identified outliers were kept for further predictive and clustering analysis as the data fell within realistic but high numbers.

Correlation Heatmap



This next visualization is a correlation heatmap which will examine potential relationships among all of the predictor variables and which could potentially influence heart disease rate.

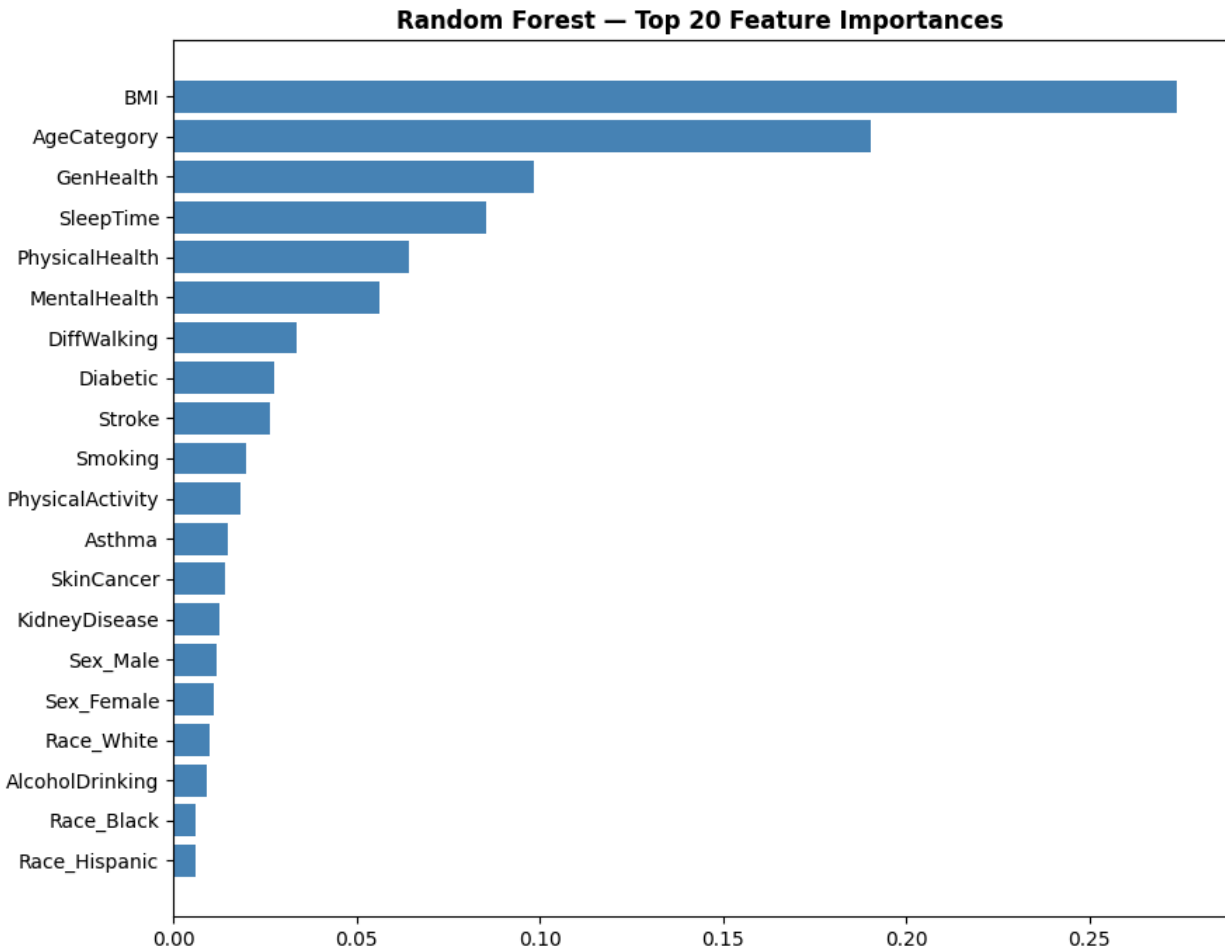
RQ1:

Logistic Regression - Odds Ratio



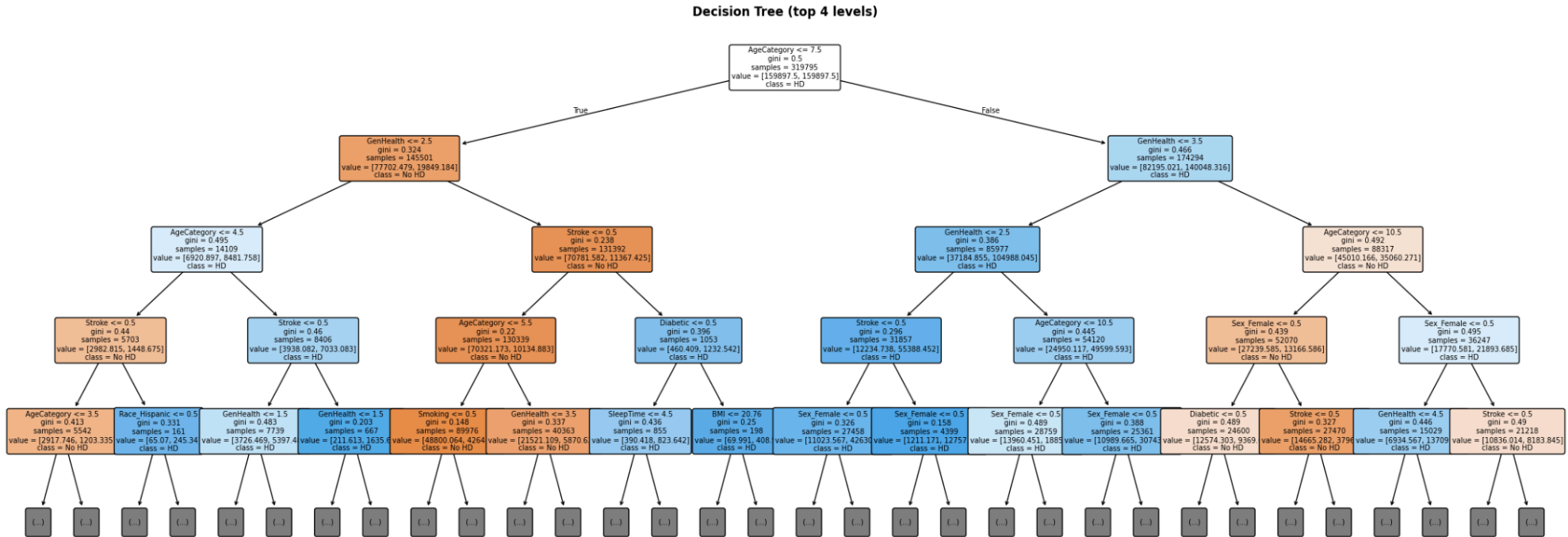
Our first model, Logistic Regression - Odds Ratio, identified which factors are the strongest positive predictors of heart disease. The Odds Ratio line is the dotted vertical line on the 1.0 value and helps visually depict which variables would increase or decrease the odds of heart disease. Variables such as Stroke, KidneyDisease, Diabetic, Smoking, Asthma, AgeCategory, DiffWalking and SkinCancer are the strongest positive predictors in that order. Stroke history has the most substantial odds associated with heart disease occurrence.

Random Forest Feature Importance



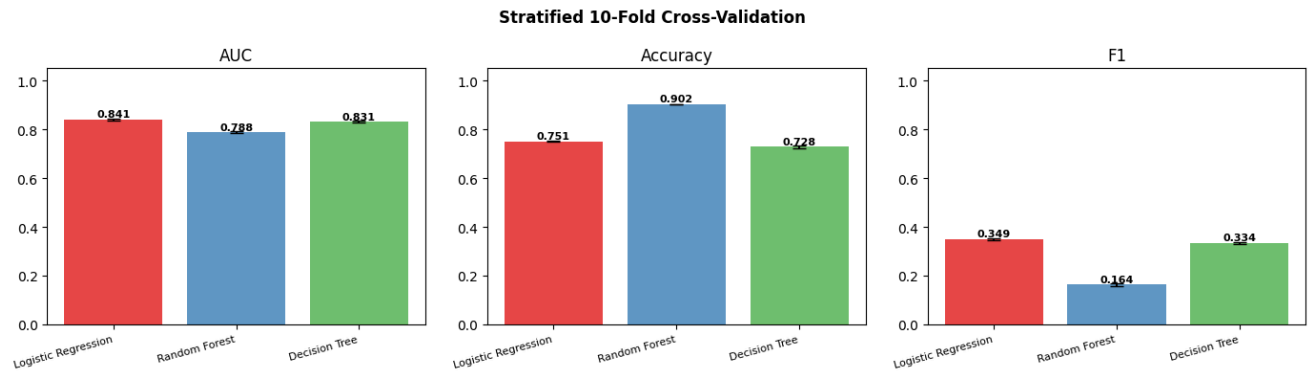
Random Forest is our second model and displayed above is the top 20 feature importances which indicated BMI, AgeCategory, GenHealth, SleepTime, PhysicalHealth and MentalHealth as the most influential predictors of heart disease. These results are quite different from our Logistic Regression model as BMI emerged as the most important predictor of the Random Forest model whereas it previously had a relatively small odds ratio. However, AgeCategory and GenHealth appeared as important predictors across both models thus far.

Decision Tree



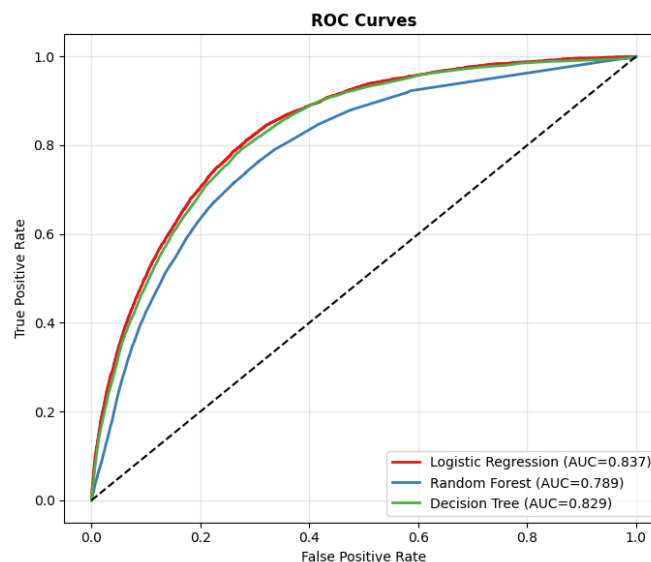
Our third model, the Decision Tree, identified AgeCategory as the primary splitting variable which is the most influential factor when determining heart disease occurrence. The rest of the splits go in this order of second-most influential and downwards: GenHealth, Stroke, Diabetic and then Smoking. This model also confirms that AgeCategory and GenHealth are consistently the most influential predictors of heart disease. Several other chronic health conditions such as Stroke and Diabetic are also influential predictors across the three models.

Stratified 10-Fold Cross-Validation



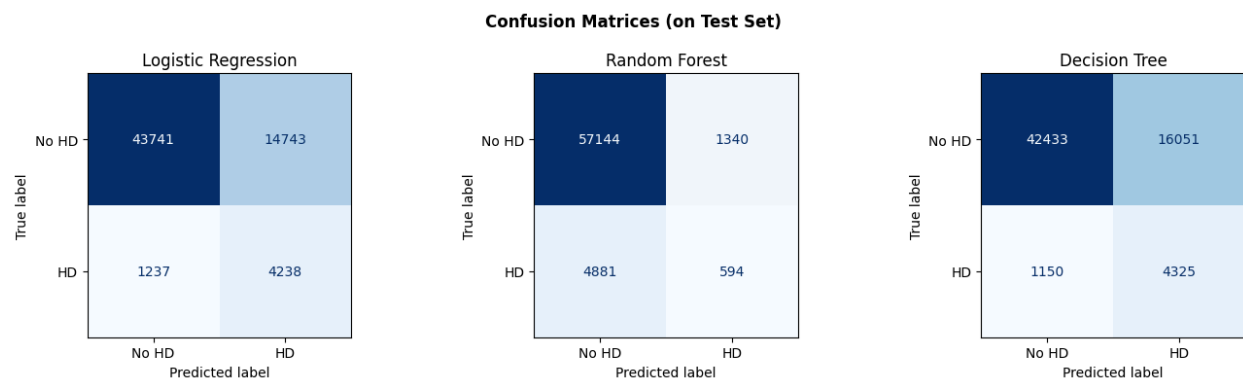
For this visualization, we combined our previous three models in order to evaluate their performance through Area Under the Curve (AUC), Accuracy, and F1-score. Logistic regression scored the highest AUC of 0.841 and a F-1 score of 0.349 with an accuracy of 0.751. The Decision Tree showed very similar numbers with an AUC of 0.831 , an F-1 score of 0.334 and an accuracy of 0.728. Lastly, Random Forest had the lowest AUC of 0.788 and F-1 score of 0.164 despite having the highest accuracy of 0.902. These results indicate that Logistic Regression performed the best in regards to having the ability to distinguish heart disease occurrence.

ROC Curves



To visually compare them with a Receiver Operating Characteristic (ROC) curve, all three models were plotted together. This curve serves as a tool to evaluate a model's ability to distinguish individuals with or without heart disease by plotting the True Positive Rate on the Y-axis and the False Positive Rate on the X-axis. The results of the ROC curve mirrors our Stratified 10-fold Cross-Validation as both Logistic Regression and Decision Tree performed similarly whereas Random Forest had a lower AUC of 0.789.

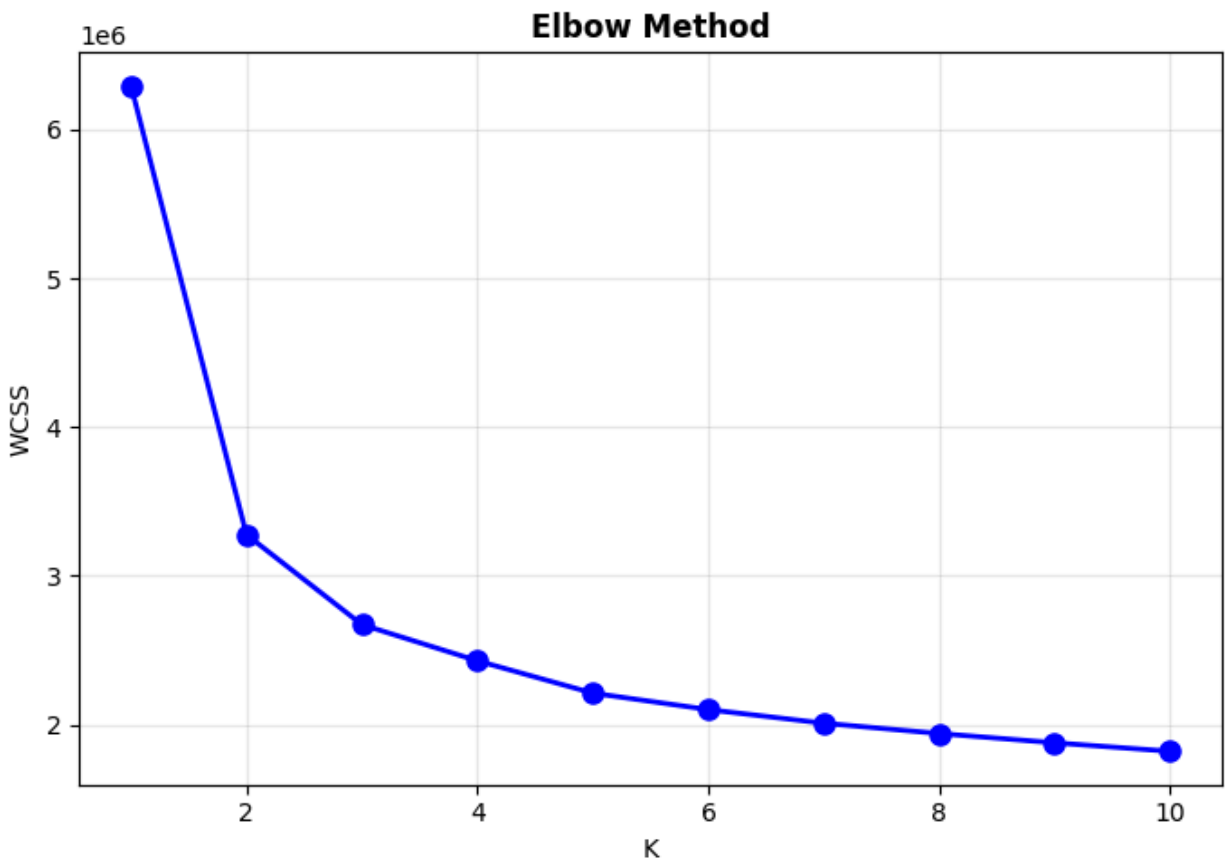
Confusion Matrices



For our last visualization for RQ1, we utilized Confusion Matrices to determine the amounts of true positives (TP), true negatives (TN), false positives (FP) and false negatives (FN). Random Forest had the largest amount of false negatives and this is reflected from a lower F-1 score from our earlier 10-fold cross-validation despite having the highest accuracy. Both Logistic Regression and Decision Tree presented a higher rate of TP and smaller rate of FP with Logistic Regression ultimately performing the best out of the three models.

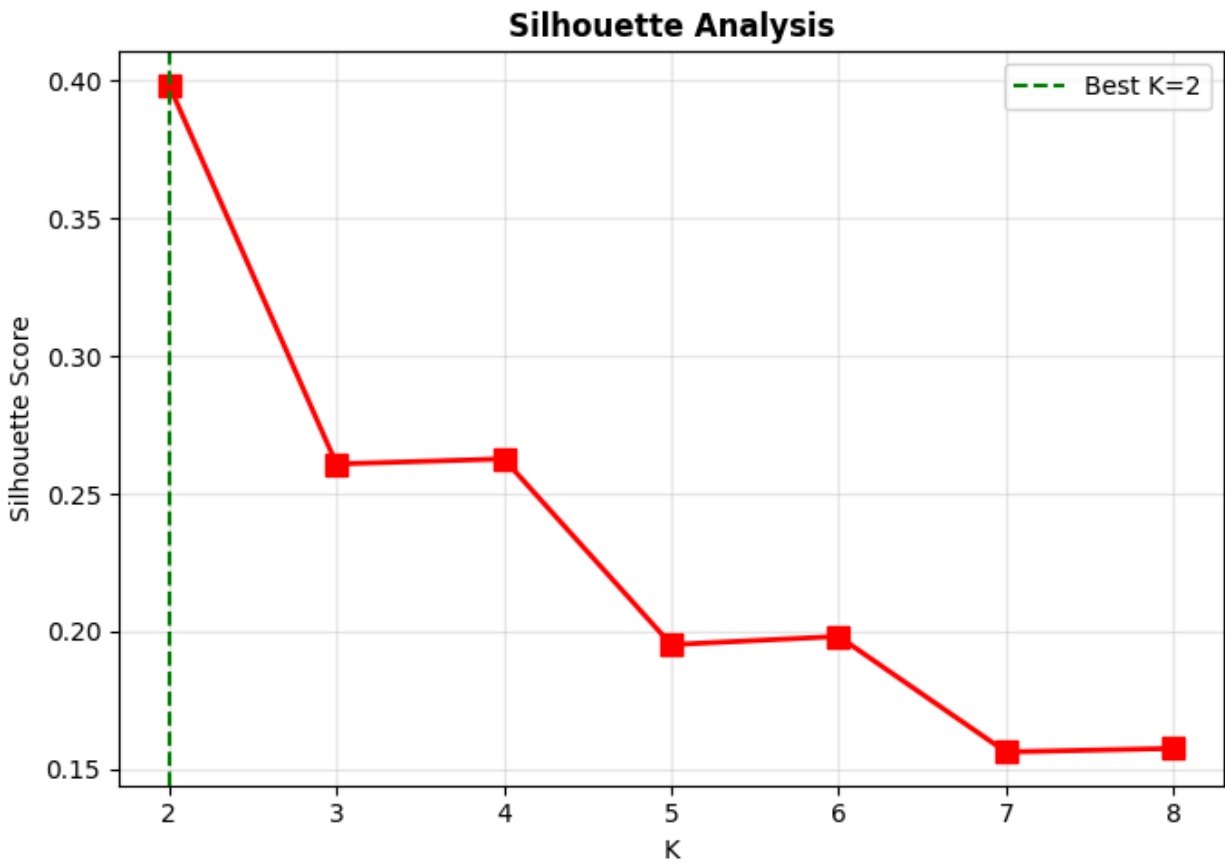
RQ2:

Elbow Method



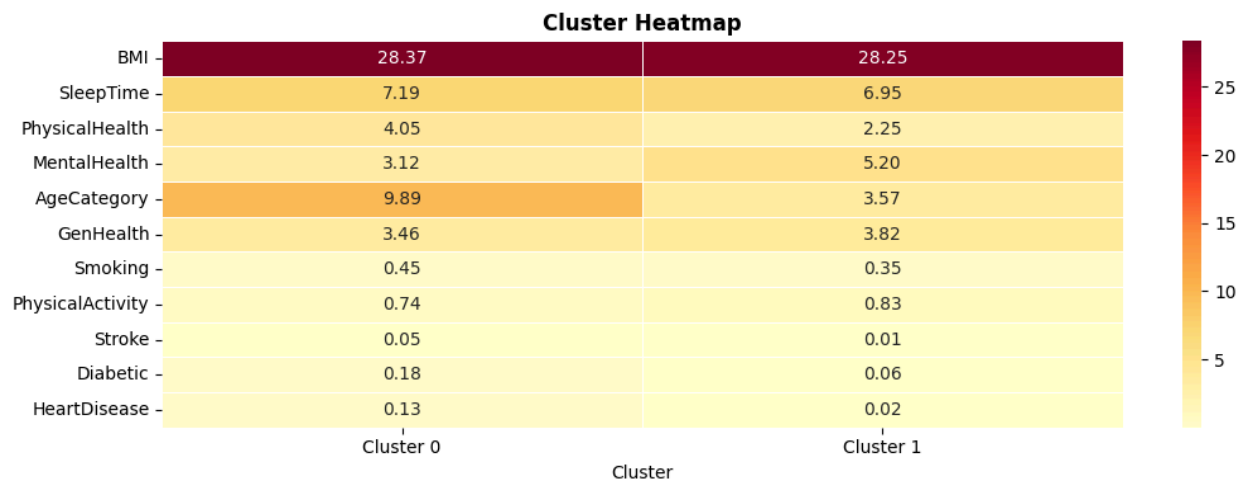
The Elbow Method was utilized to determine the optimal number of clusters (K) for the K-means clustering algorithm. The plot displays the Within-Cluster Sum of Squares (WCSS) on the Y-axis with the number of clusters (K) on the X-axis. While WCSS continues to decrease as K increases, the rate of decline begins to level off, suggesting a potential elbow point around K=2 to K=4.

Silhouette Analysis



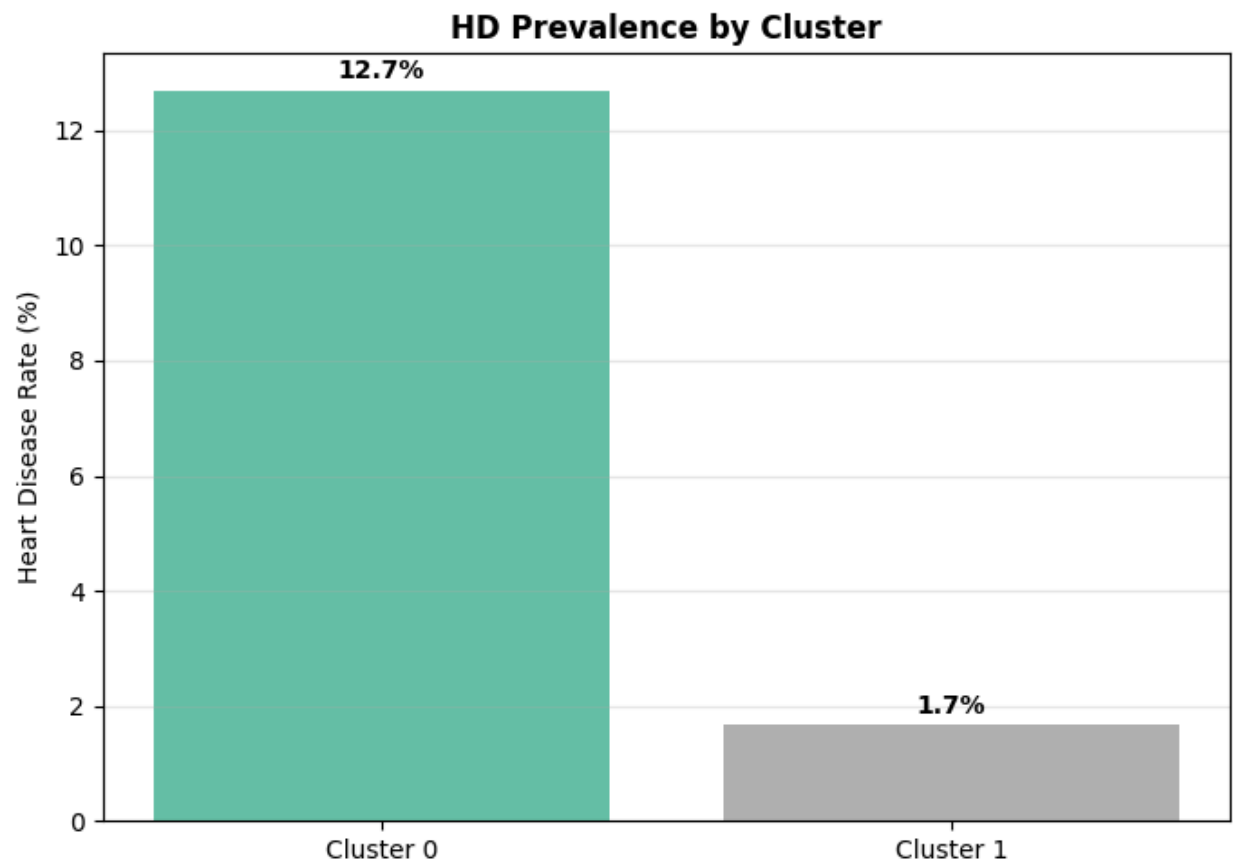
To further confirm the optimal number of clusters, a Silhouette Analysis was conducted. The highest silhouette score of approximately 0.40 was achieved at $K=2$, indicating that $K=2$ produced the most well defined and distinct clusters. As K increases beyond 2, the silhouette scores decline steadily, reinforcing that two clusters best represent the natural groupings within the data.

Cluster Heatmap



The cluster heatmap displays the mean values of each variable across the two identified clusters. cluster 0 is characterized by a notably higher AgeCategory (9.89), greater PhysicalHealth unhealthy days (4.05), and a higher heart disease rate(13%). Cluster 1 reflects a younger, healthier profile with lower AgeCategory (3.57) and higher MentalHeath unhealthy

days (5.20) Heart Disease Prevalence By Cluster



The Heart Disease prevalence by cluster visualization clearly illustrates the disparity and heart disease rates between the two clusters. Cluster 0 had a heart disease prevalence of 12.7% compared to just 1.7% in Cluster 1, suggesting that Cluster 0 represents a significantly higher risk population group.

Chi-Square Test

	χ^2	p-value
Smoking	3084.40	0.000
Stroke	4090.13	0.000

Physical Activity	3077.92	0.000
Asthma	669.98	0.000
Kidney Disease	3250.25	0.000
Diabetic	10624.09	0.000
Heart Disease	11646.81	0.000

Chi-square tests were conducted on the categorical variables to determine whether statistically significant differences existed between the two clusters. All variables including Smoking, Stroke, Physical Activity, Asthma, Kidney Disease, Diabetic, and Heart Disease returned p-values of 0.000, confirming that the cluster assignments are meaningfully associated with each of these health conditions.

ANOVA

	F	p-value
BMI	23.46	0.0000
SleepTime	2127.74	0.0000
PhysicalHealth	3886.28	0.0000
MentalHealth	5211.61	0.0000

ANOVA tests were performed on the continuous variables to assess whether differences in means across the two clusters were statistically significant. BMI ($F=23.46$), SleepTime ($F=2127.74$), PhysicalHealth ($F=3886.28$), and MentalHealth ($F=5211.61$) all yielded p-values of 0.000, confirming highly significant differences between clusters across all continuous variables.

Analysis

Model Description

Prior to our analysis, we also looked at outliers, inter-quartile-ranges (IQR) and correlation heat map as part of an exploratory data analysis to examine the distributions and quality of our data. Histograms were created in Figure 1 to look at our continuous variables including BMI, sleep time, physical health and mental health days. In addition, we also conducted IQR analysis to determine outliers as seen in Table 1. While many observations were identified as outliers, we have determined that these values are both medically plausible and representative of real-world patient symptoms and conditions and therefore were kept to preserve the integrity of this medical dataset. A correlation heatmap, as seen in Figure 3, was generated to examine the potential relationships between all of our variables.

We performed three different analyses in the form of logistic regression, random forest and decision tree for our first research question and two model analysis methods for our second research question via elbow method and silhouette analysis. RQ1 utilized 10-fold cross-validation to compare and contrast the three models whereas for RQ2, chi-square testing and Analysis of Variance (ANOVA) helps us validate our K-Means clustering analysis.

In order to evaluate the predictive performance of RQ1 via stratified 10-fold cross-validation, the dataset was split into ten equal subsets. Each of the ten subsets served as a testing set and the other nine subsets were utilized for training sets. The process was repeated ten times allowing each subset to function as a testing set allowing us to get ten folds worth of performance metrics such as AUC, accuracy and F-1 score. We then took the average of these ten folds' scores in order to compare the model performance on unseen data. Furthermore, we

generated confusion matrices utilizing a 80/20 train-test split as a holdout test set for our three models in order to visualize true positives, false positives, true negatives and false negatives.

For RQ2, we utilized two models to determine the optimal number of clusters for K-Means clustering in order to identify patient risk profiles based on demographic, behavioral, and health-related characteristics. Following our cluster analysis, chi-square tests of independence were conducted on the categorical variables such as the statuses of smoking, stroke, asthma, kidney disease, diabetes, heart disease and physical activity. ANOVA was conducted on the continuous variables such as BMI, sleep time, days of bad physical health and days of bad mental health.

Model Results

RQ1 - Which demographic, behavioral, and health-related factors are the strongest predictors of heart disease among U.S. adults?

Across all three models, 'Age Category,' 'General Health,' 'Diabetes,' and 'Stroke' were identified as significant predictors. The logistic regression model showed health related factors such as skin cancer, diseases, and, most of all, history of a stroke to be the most significant factors in predicting cardiovascular disease as seen in Figure 6. The random forest model, in Figure 7, identified a person's BMI to be the most significant factor in predicting the risk of cardiovascular disease, followed by 'Age Category,' 'General Health,' and 'Sleep Time.' Figure 8 shows how the decision tree model identified 'Age Category' as the prime splitting variable, which indicates it is the most important variable when predicting the risk of heart disease. The results also suggested 'General Health' and 'Stroke' to be highly significant.

The performance of the models were evaluated using stratified 10-fold cross validation. Logistic regression had the highest performance with an AUC of 0.841, F1-score of 0.349, and

accuracy of 0.751 as seen in Figure 3. The decision tree produced similar results. The random forest resulted in the highest accuracy (0.902), but the lowest AUC (0.788). According to these results, the most significant factors when predicting heart disease are ‘Stroke,’ ‘Age Category,’ and factors related to ‘General Health.’

Furthermore, we evaluated the three models by plotting their ROC Curves in one graph as seen in Figure 4. All of these three models showed predictive capability substantially above random classification (the diagonal reference line). Next, we also included confusion matrices in Figure 4. The logistic regression model correctly classified 437,741 non-heart diseases cases and 4,238 heart diseases cases with 14,734 false positives and 1,237 false negatives. The decision tree model correctly predicted 42,433 non-heart disease cases and 4,325 heart disease cases with 16,051 false positives and 1,150 false negatives. The random forest model differs from the previous two models as it correctly predicted 57,144 heart disease cases and 594 non-heart disease cases with 4,881 false positives and 1,340 false negatives. While the random forest model had a higher overall accuracy than the other two models, it also produced the most false negatives. This confusion matrix supports the results from our ROC curve analysis where logistic regression showed the best performance in having the highest AUC score and a stronger balance between minimizing errors and correctly predicting heart disease cases.

RQ2 - What common characteristics define patient risk profiles and how do heart disease rates vary amongst these groupings?

We utilized K-Means clustering in order to determine patient risk profiles and afterwards, we applied the elbow method and conducted a silhouette analysis to identify the optimal number of clusters. The curve in Figure 9 shows that the curve begins to flatten at $K = 3$ which suggests that higher clusters have diminishing improvements. In order to validate this, we also calculated

the R^2 and additional R^2 values in Table 1. These results show that additional R^2 going from $K=2$ to $K=3$ adds 9.61% variance, from $K=3$ to $K=4$ adds 3.86% variance, from $K=4$ to $K=5$ adds 3.45% variance and going from $K=5$ to $K=6$ adds 1.76% variance. This pattern indicates that the clustering stabilizes at around $K=5$ to $K=6$. Silhouette analysis identified the optimal number of clusters at $K = 2$ which has the highest silhouette score of 0.3984 as seen in Figure 10 and Table 2. We ultimately determined that two clusters should be utilized for our following analysis.

Cluster 0 was characterized by an average age category of 9.89, higher rates of heart disease at 0.13, diabetes at 0.18, stroke at 0.05 and physical health at 4.05. On the contrary, Cluster 1 was characterized by an average age category of 3.57, lower rates of heart disease at 0.02, diabetes at 0.06, stroke at 0.01 and physical health at 2.25. BMI results were nearly identical between the two clusters with values of 28.37 and 28.25 for the first and second clusters, respectively.

Figure 11 depicts the heart disease prevalence between the two clusters. Cluster 0 showed a 12.7% heart disease prevalence rate whereas Cluster 1 exhibited a 1.7% heart disease prevalence rate. The observed differences indicate the substantial variance of heart disease occurrence between the two clusters utilized for K-Means cluster analysis.

Figure 12 illustrates a cluster heat map for our two selected clusters. Cluster 0 had values of AgeCategory at 9.89, SleepTime at 7.19, PhysicalHealth at 4.05, GenHealth at 3.46 and MentalHealth at 3.12. On the other hand, cluster 1 had values of SleepTime at 6.95, MentalHealth at 5.20, GenHealth at 3.82, AgeCategory at 3.57 and PhysicalHealth at 2.25. BMI for both clusters exhibited similar values at 28.37 and 28.25 for Clusters 0 and 1, respectively.

Results

RQ1 - Which demographic, behavioral, and health-related factors are the strongest predictors of heart disease among U.S. adults?

Through our three models of logistic regression, random forest and decision tree, the findings indicate that AgeCategory, GenHealth, Diabetes and Stroke were the strongest predictors of heart disease among U.S. adults. Factors such as older age, poorer general health, being diabetic and having a history of stroke were closely associated with an increased likelihood of heart disease. Despite variables such as BMI being indicated by specific individual models, AgeCategory, GenHealth, Diabetes and Stroke presented the best consistency across all three modeling analyses. These findings support our hypothesis that these specific predictors contribute significantly to cardiovascular disease risk.

RQ2 - What common characteristics define patient risk profiles and how do heart disease rates vary amongst these groupings?

Through clustering analysis, we identified two distinct patient risk profiles where cluster 0 represented a higher-risk patient profile characterized by older age, higher rates of heart disease, diabetes, stroke and poorer physical health resulting in greater cardiovascular risk. The second cluster represents a lower-risk patient profile characterized by younger age, lower rates of heart disease, diabetes, stroke and better physical health resulting in lower cardiovascular risk. The higher-risk cluster exhibited a much higher prevalence of heart disease at 12.7% compared to the lower-risk cluster being at 1.7%. These findings help indicate that significantly different risk profiles exist within our dataset population and that heart disease prevalence varies drastically amongst these groups.

Ethical Recommendations

Healthcare is a very serious field as it deals with individuals' wellbeing daily. For this reason, there is very little room for error, whether it is in the operating room, regular check-ups, or making the proper diagnosis. The lives of real people are at risk if mistakes are made, which highlights how important it is to build healthcare related algorithms with precision, accuracy, and understanding of potential biases. By developing an accurate and ethical predictive model that identifies individuals who are at risk of developing cardiovascular disease, healthcare providers can intervene before serious health issues develop.

Predictive models based on real, diverse datasets have the potential to support and improve preventative healthcare and diagnosis procedures. The quality and reliability of the data is a great concern to the ethics of incorporating machine learning into healthcare. A systematic review on the risk of bias in predictive modeling has shown around 40% of models handle missing or inaccurate data inadequately (Andaur Navarro et al., 2021). The data used in this analysis is self-reported, so it is important to take care in looking for discrepancies or falsities in the data. Personal bias and reporting errors are common issues with self-reported data, which can lead to false general health, smoking, or drinking statuses.

The consequences of prediction errors or security breaches are great when it comes to predicting health concerns. False positives can cause unneeded stress and medical costs, and false negatives may delay prevention care and lead to more serious cardiovascular issues. While we strive for accuracy when developing a predictive model, this cannot replace a professional diagnosis. In addition to this, privacy and data security is vital when working with such sensitive

data. Unethical use of this data may result in biases, discrimination, or misuse of the data if the proper precautions are not taken.

Fairness and bias are a serious issue in the healthcare community. Data that has variables such as race, age, and sex comes with the potential of accidentally reinforcing existing healthcare stereotypes if not analyzed carefully. Wang et al. (2024) found that racial bias in predictive modeling is quite complex, and different models produce differing levels of bias. Since a model may be better at classifying one race or gender than another, these incorrect diagnostics can further healthcare disparities. However, including variables like these can improve the model's performance. Predictive models should always be evaluated with fairness across all demographics to ensure equitable predictions.

In addition to racial bias, another fairness concern for cardiovascular disease prediction is sex-based or gender-based bias. Li et al. (2023) evaluated machine learning models for cardiovascular disease prediction and found that these models can produce unfair differences in performance across different demographic groups. More specifically “ML models were biased against women, suggesting the need to investigate and correct gender disparities in CVD risk prediction” (Li et al, 2023). This is important because a heart disease model that predicts risk more accurately for one sex than another could lead to unequal prevention, diagnosis, or treatment recommendations.

Despite the ethical concerns, the responsible use of predictive analytics has the potential to improve the health of the community. Identifying high-risk individuals sooner rather than later can reduce healthcare expenses and improve patient outcomes. When used ethically and responsibly, predictive models can serve as useful tools in the prevention of cardiovascular disease. Predictive analytics should be used in support of healthcare, not replace it. By

prioritizing fairness, equity, and privacy, predictive analytics can help prevent serious cardiovascular disease.

Challenges

Some challenges that were faced with our dataset :

- Outliers
 - A challenge we encountered during our analysis was determining how to handle outliers within the variables. Using the Interquartile Range method, several observations were flagged as potentially outliers. However, in the context of health-related data, these values are not necessarily errors. For example, a respondent reporting 30 days of poor physical health within a month may represent a severe but realistic health condition rather than an invalid data point. Because of this, removing all flagged outliers would have created a misleading dataset by excluding individuals with more serious health issues, which may be especially important in a heart disease prediction project because poor physical or mental health may be connected to higher cardiovascular risk. Therefore, instead of automatically removing these observations, we kept them in the dataset because they were medically plausible and represented real-world variation among patients.
- Uneven race distributions
 - Another challenge we encountered was the uneven distribution of race within the dataset. The dataset contained individuals from several racial groups, but the majority of respondents were White, making up approximately 76.7% of the dataset. Because of this imbalance, the model may be better at identifying patterns for the largest racial group than for smaller racial groups with fewer observations. This causes the raw counts to be misleading, for example, white

respondents had the largest number of heart disease cases, but this is partly because they also made up the largest portion of the dataset. The uneven race distribution may also affect the performance and fairness of the predictive models. If some racial groups are underrepresented, the model may not learn enough information about those groups, which could lead to less accurate predictions for them. This is especially important in a health-related predictor because biased predictions could lead to unfair or unreliable risk estimates.

Recommendations

In this project, we used several health, lifestyle, and demographic variables to predict whether a person may have heart disease. These included factors such as BMI, smoking, alcohol drinking, stroke history, physical health, mental health, difficulty walking, sex, age category, race, diabetes, physical activity, general health, sleep time, asthma, kidney disease, and skin cancer. While these variables gave us useful information, there are several ways the analysis could be improved in the future.

One recommendation would be to include more direct medical measurements. Many of the variables in this dataset are general health indicators. If we were to do this project again, we could improve the model by using a dataset that would include more clinical factors such as blood pressure, cholesterol levels, resting heart rate, medication use, and family history of heart disease. By including these variables we could give a more complete and accurate picture of a person's risk for heart disease.

Next steps should build on the results from both research questions by improving the model performance, strengthening cluster validation, and making the findings easier to apply in a real-world health setting. A good real-world adjustment would be to focus more on reducing false negatives. These are especially important to avoid in a real world application because predicting that someone does not have heart disease when they do, could be more serious than even a false positive. Since heart disease cases are less common than non-heart disease cases, future analysis should address possible class imbalance by using methods like class weighting, which could help with reducing the number of false negatives.

Another important step would be to validate the clusters using a separate sample or repeated random samples. If the same high-risk and low-risk profiles appear consistently, that

would provide stronger evidence that the clusters represent meaningful patterns in the data. A three and four cluster solution could also be tested to see if additional clusters reveal more specific patient risk profiles than the two cluster solution that was used. This could help move beyond a higher versus lower risk grouping to potentially identify more detailed subgroups based on combinations of age, diabetes, stroke history, physical health, mental health, and lifestyle factors.

Overall, this project showed that heart disease prediction can be improved by combining predictive modeling with cluster analysis to identify both individual risk factors and broader patient risk profiles. This suggests that variables such as age, general health, diabetes, stroke history, and physical health are important in understanding heart disease risk. However, future analysis could be strengthened by using more direct clinical measurements, addressing potential class imbalance, and focusing on reducing false negatives. The results would become more reliable, interpretable, and useful for identifying groups of individuals who may be at higher risk for heart disease.

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Appendix

Figure 1.

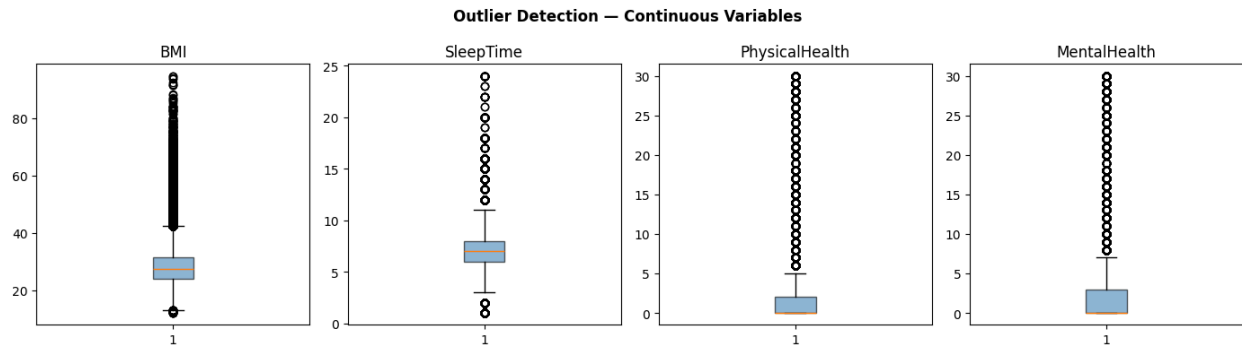


Figure 2.

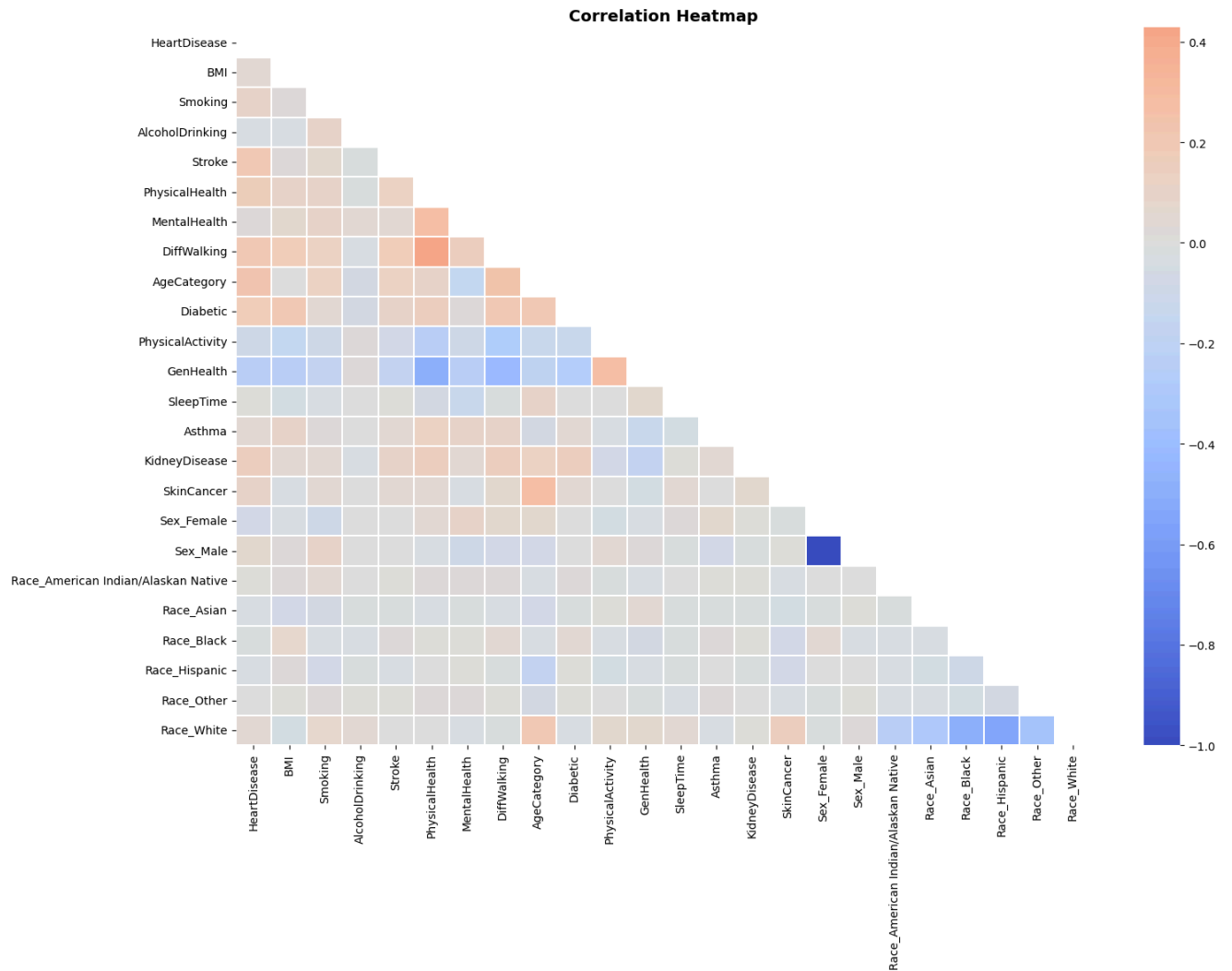


Figure 3.

Stratified 10-Fold Cross-Validation

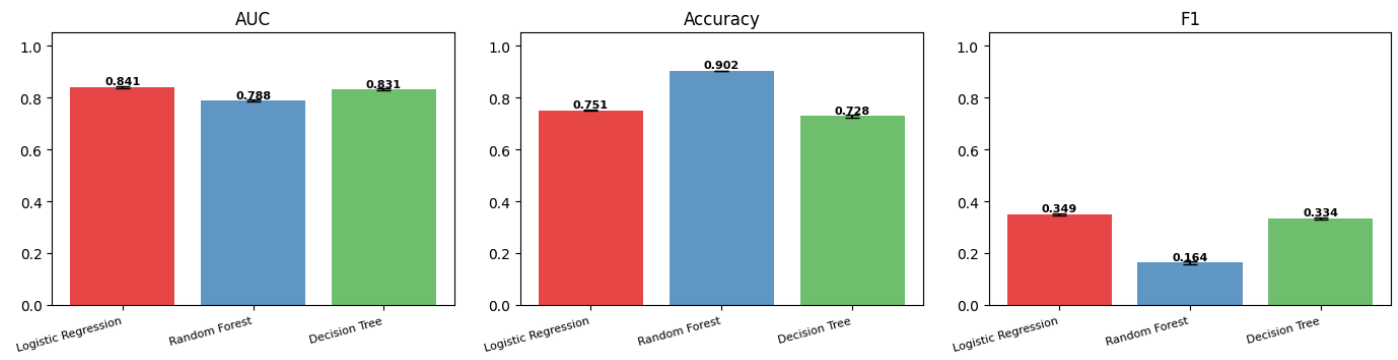


Figure 4.

Confusion Matrices (on Test Set)

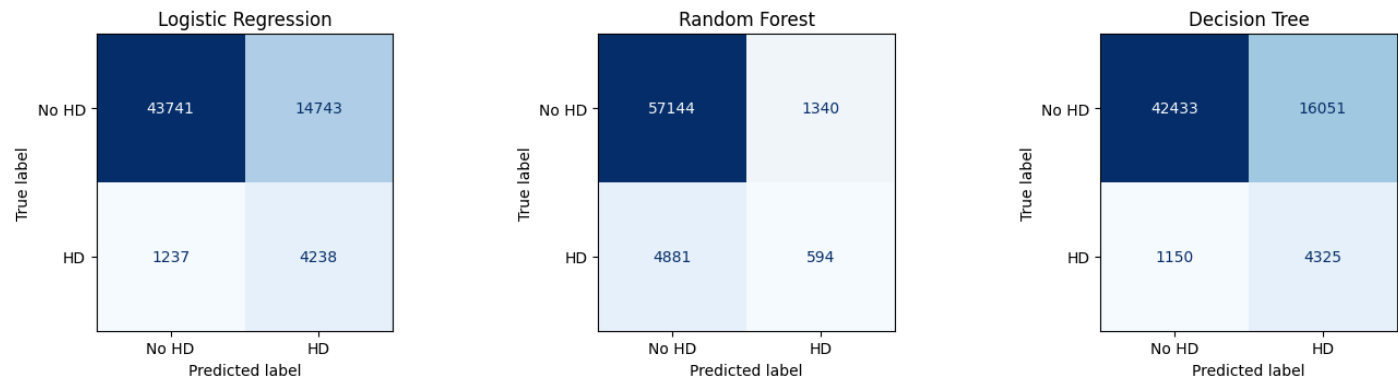


Figure 5.

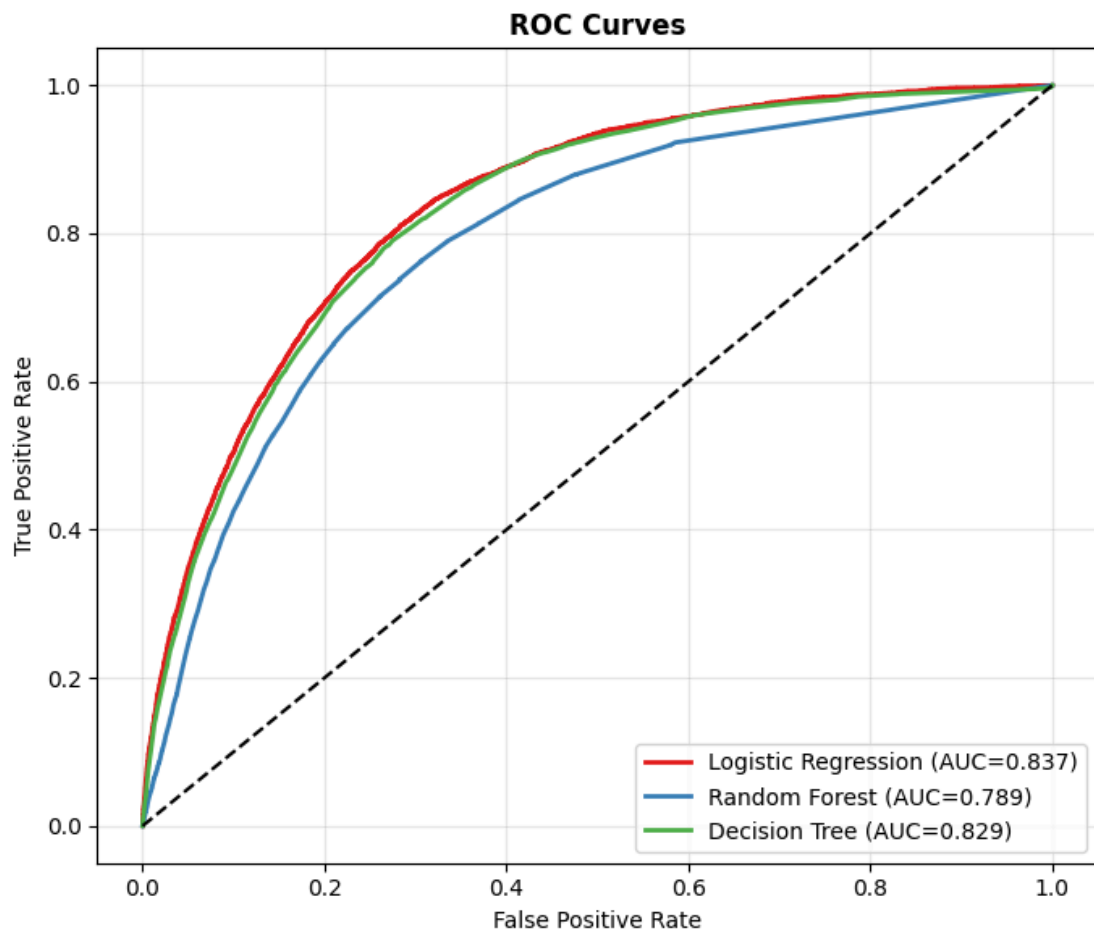


Figure 6.

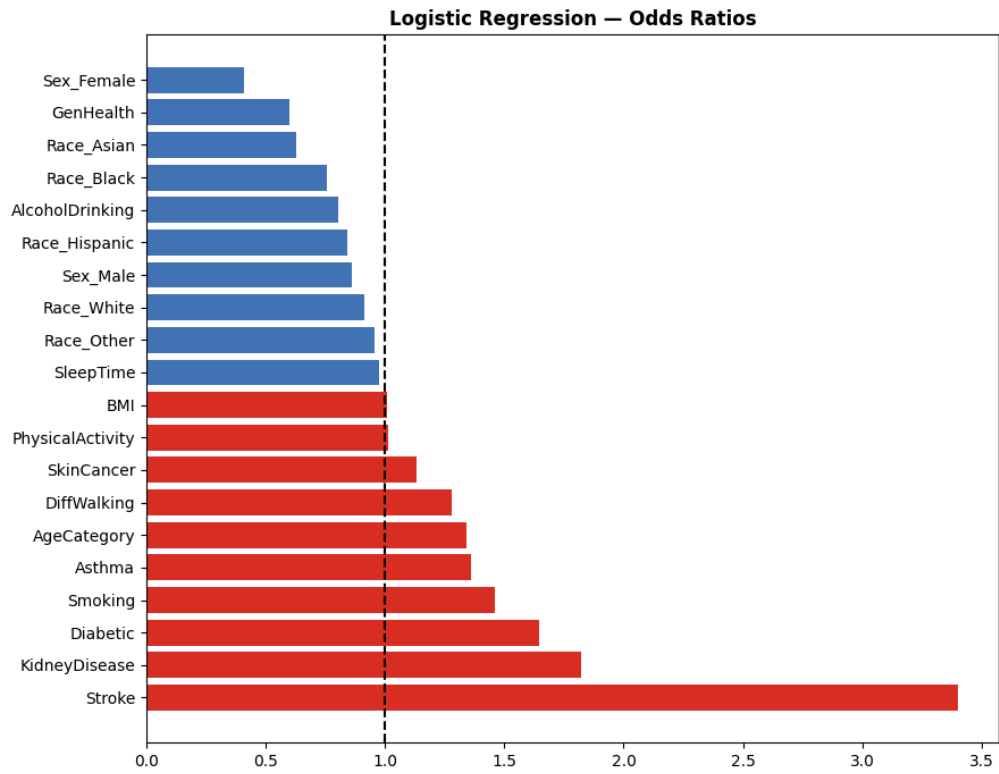


Figure 7.

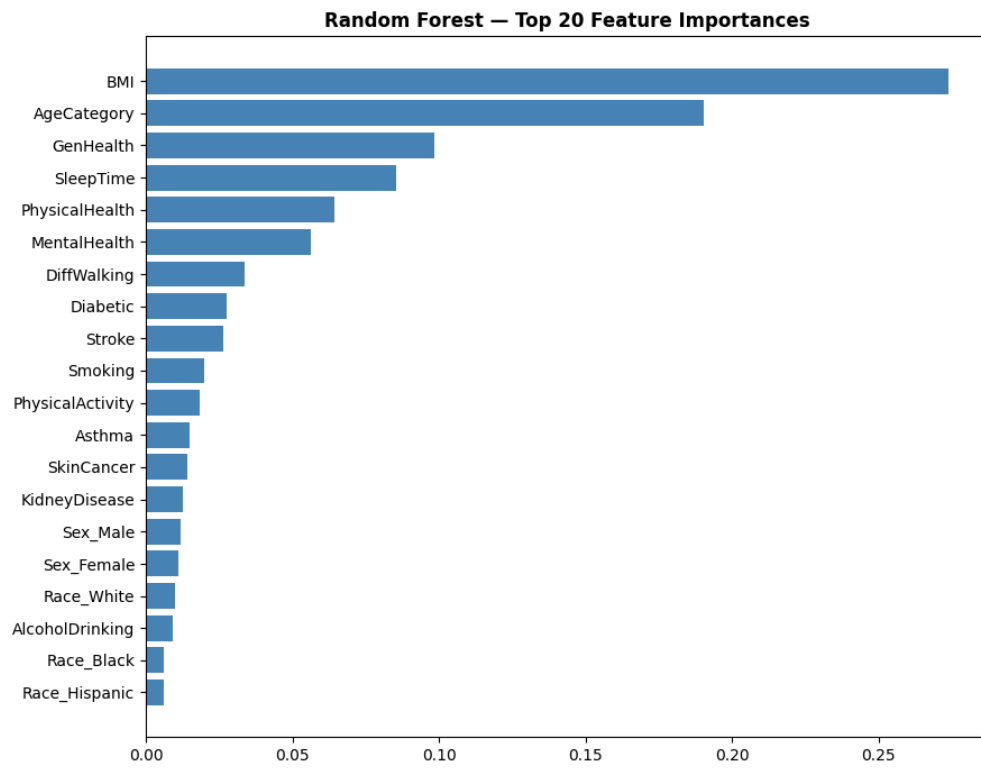


Figure 8.

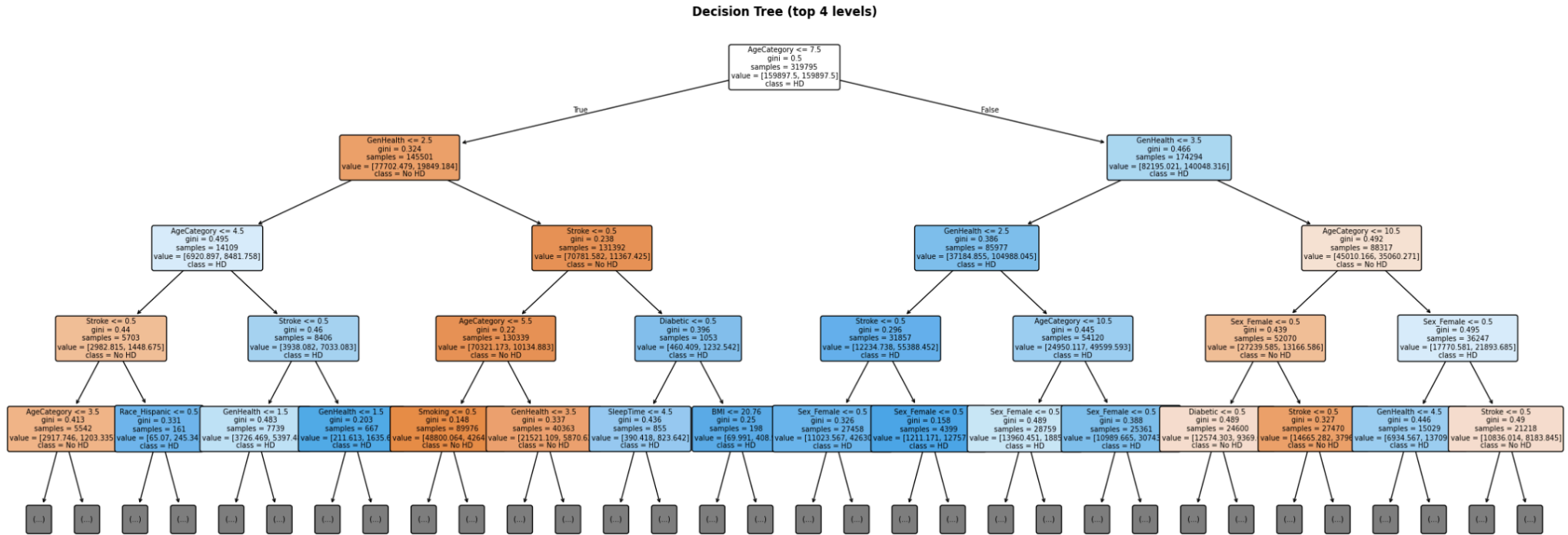


Figure 9.

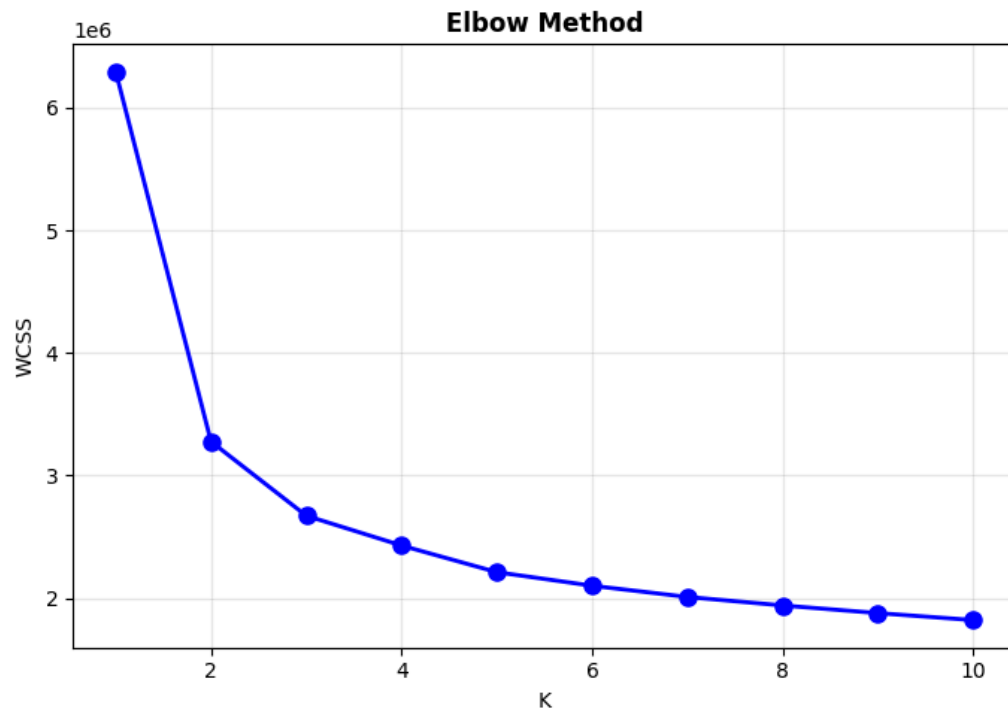


Figure 10.

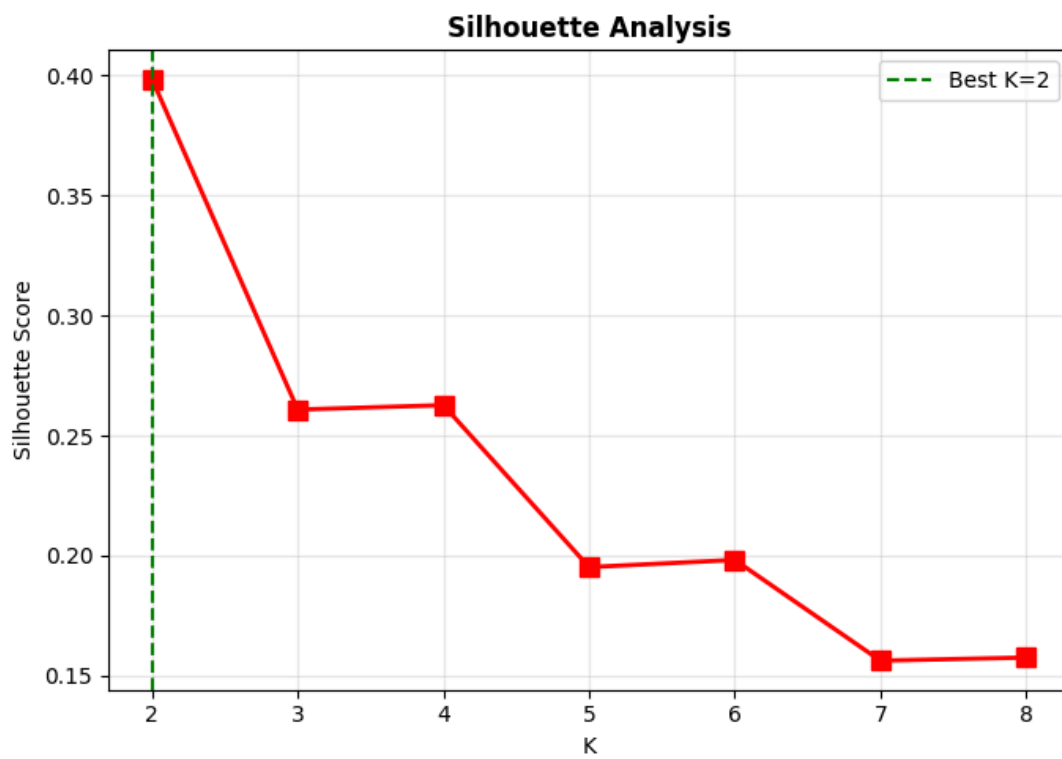


Figure 11.

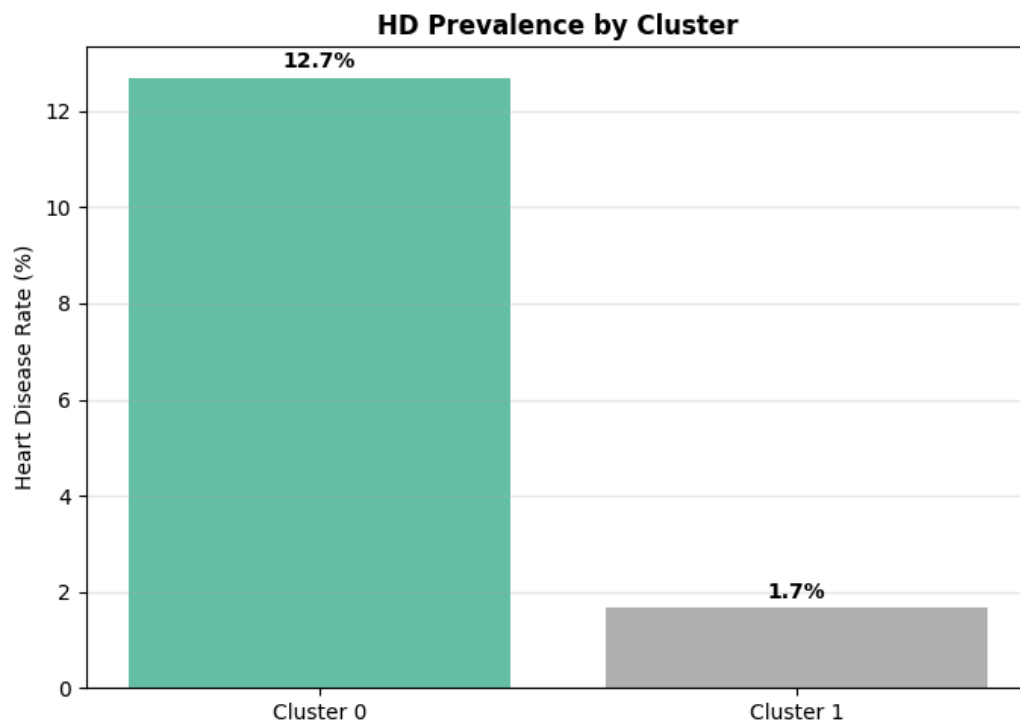


Figure 12.

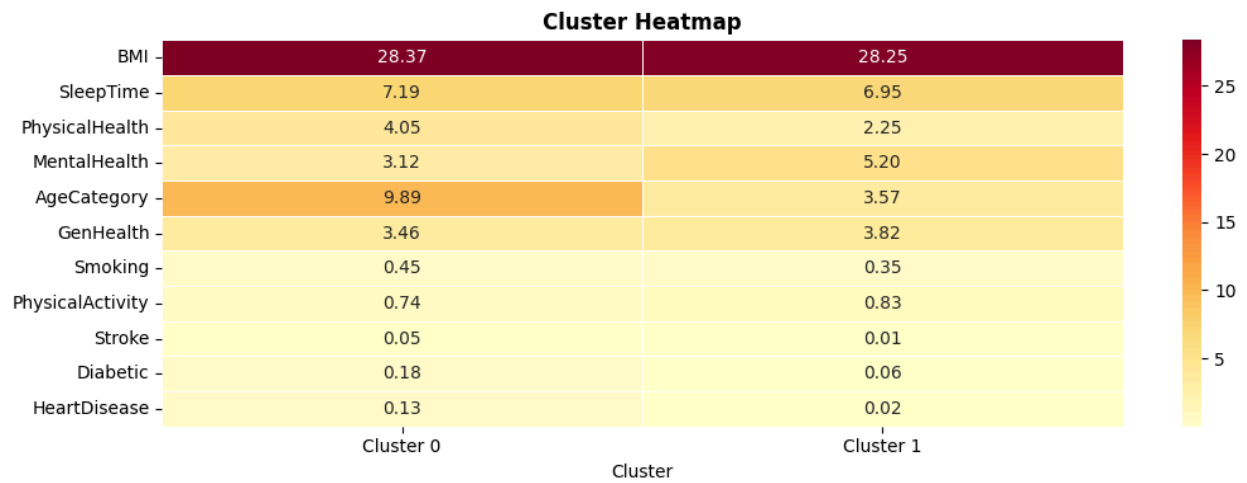


Table 1.

	K	WCSS	R ²	Additional R ²
0	1	6.293753e+06	-0.0000	NaN
1	2	3.276278e+06	0.4794	0.4794
2	3	2.671335e+06	0.5756	0.0961
3	4	2.428112e+06	0.6142	0.0386
4	5	2.210760e+06	0.6487	0.0345
5	6	2.100175e+06	0.6663	0.0176
6	7	2.008540e+06	0.6809	0.0146
7	8	1.939335e+06	0.6919	0.0110
8	9	1.876893e+06	0.7018	0.0099
9	10	1.819850e+06	0.7108	0.0091

Table 2.

Number of Clusters	Silhouette Score
K = 2	0.3984
K = 3	0.2608
K = 4	0.2627
K = 5	0.1952
K = 6	0.1982
K = 7	0.1563
K = 8	0.1576

Table 3.

	IQR Bounds	Outliers Flagged
BMI	[12.9, 42.5]	10396
SleepTime	[3.0, 11.0]	4543
PhysicalHealth	[-3.0, 5.0]	47146
MentalHealth	[-4.5, 7.5]	51576

Source Code

<https://github.com/legume-lover/DAT490CapstoneProject>